

The Homelessness Trap: Labor Market Trajectories of Individuals Experiencing Homelessness

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September 19, 2026

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Abstract

Debates around reducing homelessness point to the role of labor markets in providing an income to afford rent and thus a roof. Yet a rising number of individuals become homeless while working or having recently worked. I study whether job loss causes homelessness and whether homelessness, in turn, raises job separation and lowers job finding. To do so, I link the universe of homeless shelter requests in France to administrative employment records, covering over one million individuals. To identify each direction of the cycle, I use a matched difference-in-differences design around economic layoffs and an instrumental variables design based on the leniency of call-center agents allocating shelter. I find that job loss raises the probability of homelessness by 50 percent over three years, concentrated after unemployment benefits expire. In the reverse direction, obtaining a shelter bed at the first request reduces the risk of job separation within three months by 17 percent and raises the probability of being employed one year later by 74 percent. Unemployment and homelessness thus reinforce each other, and rapid shelter access breaks the cycle.

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1 Introduction

Homelessness has been rising in many advanced economies. By the end of 2024, the United States recorded approximately 770,000 unhoused individuals ([Porter, 2024](#)) and France roughly 350,000 ([Dihal, 2023](#)). Reducing homelessness has thus moved to the forefront of social policy. Labor market income is typically seen as the prerequisite for affording rent and thus a roof. This raises the question of the role of employment in both reducing entry and accelerating exits out of homelessness. Yet, while the causes and consequences of unemployment have been studied extensively, we have little knowledge of the role of labor markets in explaining homelessness. As labor market trajectories of unhoused individuals are hard to observe, the extent to which job loss and unemployment are a cause of homelessness, a consequence of it, or both remains unclear. Additionally, the two might reinforce each other causing a vicious cycle between unemployment and homelessness.

The extent to which a vicious cycle between homelessness and unemployment exists has been hard to establish. It requires panel data identifying individuals experiencing homelessness and their labor market situation over time, as well as sources of quasi-random variation to identify causal effects. Entries and exits out of homelessness are difficult to observe, as there are rarely any data sources capturing this population. Surveys typically only capture a fraction of the homeless population at one moment in time and tend to oversample long-term homeless ([O’Flaherty, 2019](#)). Even with the most recent data advances ([Costantini, 2026](#); [Meyer, Wyse, Grunwaldt, Meyer and Wu, 2023](#)), we rarely know what has happened before and after the moment an individual was identified as homeless. Furthermore, given that job loss and housing loss are often correlated, we require exogenous variation in both directions of the vicious cycle to estimate its effects credibly. Yet, existing data sources provide little to no quasi-random variation that can be exploited.

In this paper, I examine the vicious cycle between homelessness and unemployment tracking over one million individuals experiencing homelessness in France over time. To test whether this cycle exists, I estimate its causal effects in both directions. First, I examine the effect of labor market shocks on homelessness. Then, I investigate the reverse channel of housing shocks on labor market trajectories.

My empirical exercise is set in France, which offers an informative setting to examine whether a vicious cycle between homelessness and unemployment exists. First, homelessness

per capita is among the highest in advanced economies. The number of individuals experiencing homelessness rose from 143,000 in 2012 to 350,000 in 2025, or 0.5% of the population, compared with an average of 0.25% across OECD countries ([OECD, 2024](#)). Second, most entries into and exits from homelessness can be tracked through the shelter system, whose requests and stays are recorded in a nationwide management system, *SI-SIAO*. An estimated 95% of individuals experiencing homelessness in France pass through this system at some point. Requests for emergency shelter are made through a national hotline, *115*, which in 2025 received requests from between 25,000 and 30,000 distinct individuals per week. This number of requests far exceeds the places available: in 2025, 60% of requests went unmet in a given week on average.

I construct the largest database to date of individuals experiencing homelessness, combining nationwide shelter records with individual-level administrative employment and unemployment data. The two sources are merged based on names and date of birth in a secure environment. The shelter records identify every request, independent of its outcome, and every stay over time. The administrative employment and unemployment data allows me to recover precisely what happened before, during, and after the homelessness spell. As a result, I can study the labor market trajectories of over one million individuals who experienced homelessness in France between 2017 and 2025.

Plotting individual's labor market trajectories around the first request for shelter confirms recent evidence that many individuals experiencing homelessness are still active in the labor market ([Meyer, Wyse, Grunwaldt, Meyer and Wu, 2023](#); [Von Wachter, Schnorr and Riesch, 2020](#)). At the time of the shelter request in 2024, 8.6% of individuals were still working, and 26.7% had worked within the two years before their first request. Examining employment and earnings around the first request for those who had worked for at least six months within the previous two years, I observe a sharp drop around the time of the request. Most individuals who remain active in the labor market report very low earnings and are employed in precarious short-term contracts. Consistent with this, I find a strong negative correlation between earnings and the risk of becoming homeless in the following year; yet at every earnings level, this risk is substantially lower for workers with an open-ended contract than for those with a more precarious fixed-term contract. These patterns highlight the interdependence between an individuals' employment and housing situation, challenging the implicit assumption of job search models not accounting for a job seeker's housing situation.

I extend a [McCall \(1970\)](#) job search model to account for the housing situation of the job

seeker, and use it to study how employment and housing interact. Entry into homelessness is governed by an affordability condition: workers face exogenous housing shocks that trigger a redraw of their rent, and because they are hand-to-mouth, they become homeless whenever the new rent exceeds their current income. This condition makes unemployed workers, and in particular those receiving only welfare benefits, especially vulnerable to housing shocks. Job offer and separation rates are indexed by housing status: unhoused workers, who lack a stable address and a place to rest, may face a lower offer rate and a higher separation rate. In this case, a lower reservation wage follows: the option value of continued search falls when jobs are shorter-lived and offers scarcer, so unhoused workers accept lower-paid jobs. These are precisely the jobs least likely to clear the affordability condition, which slows their return to housing and prolongs both spells. The model thus makes the existence of the cycle contingent on two gaps that can be estimated, which the two quasi-experiments address in turn.

I first estimate the causal effect of job loss on homelessness. Using one-to-one matching, I compare workers laid off for economic reasons to similar workers employed in firms that did not undergo economic layoffs.

I find that job loss raises the risk of becoming homeless by 50% over the three years following the layoff. Restricting attention to the first shelter request, I observe that the sharpest increases in homelessness risk occur in the quarters after unemployment benefits expire, pointing to liquidity constraints. Heterogeneity analysis reinforces this interpretation: the effect is concentrated among workers who are single and earned a low income in their previous job.

To examine the potential vicious cycle, I turn to the reverse direction: the effect of homelessness on labor market outcomes. To do so, I exploit the allocation of scarce shelter places. A request for shelter is handled by a call-center agent assigned quasi-randomly within an area (*département*), month, and shift, and agents differ systematically in how often they meet a request. Because the shelter system is highly saturated, whether a first request is met determines whether an individual is sheltered immediately or spends time on the street. Thus, call center agent leniency isolates a positive housing shock at the moment of entry into homelessness. Importantly, I show that the worker's labor market history is not predictive of the call center agent's leniency. I furthermore validate the findings of this design, using a different type of housing shock, zooming in on unhoused individuals that became homeless after their landlord reclaimed their flat for sale or personal use.

Rapid shelter access matters for both job separation and job finding. Among callers em-

ployed at the time of their request, a met request reduces the probability that the contract ends within three months by 17% and reduces contract terminations, notably quits and dismissals at the end of the trial period, by a third. Among callers unemployed at the time of the request, a met request raises the probability of employment one year later by three quarters, from a base of only 8%, and raises days worked, earnings, and open-ended employment by similar proportions. A central mechanism behind these findings seems to be the reduction in exposure to street homelessness. Individuals whose first request is denied spend 50 fewer nights in shelters over the following year.¹ Consistent with these findings, housing placements have been shown to reduce homelessness and improve socioeconomic outcomes, including employment (Brounstein and Wieselthier, 2024; Cohen, 2024).

To understand the magnitude of the vicious cycle and of homelessness in explaining prolonged unemployment spells, I use the causal estimates from the two quasi-experiments to estimate the extended McCall job search model. I show XXX.... I then use the estimated model to evaluate alternative policies aiming to reduce homelessness.

This paper contributes to four strands of the literature.

First, I add to the literature on the causes and consequences of homelessness. A first set of papers studies its causes, showing that local housing and labor market conditions shape entry into homelessness (Bruneel-Zupanc, Gaillard, Hellwig, Wangner and Werquin, 2026; Quigley and Raphael, 2001), and that individual shocks, including health and mental health episodes, raise the risk of becoming homeless (Curtis, Corman, Noonan and Reichman, 2013; van Ours and Moschion, 2021). A related literature on eviction documents that job loss is among the main risk factors behind defaults on rent, and that the associated income losses are persistent (Abramson, 2024; Collinson, Humphries, Mader, Reed, Tannenbaum and Van Dijk, 2024). In these settings, however, the object of study is the eviction: job loss characterises the population that reaches default, and its effect on homelessness is not separately identified. I fill this gap by providing the first causal estimate of the effect of job loss on homelessness. A second set of papers studies the consequences of homelessness (Cobb-Clark and Zhu, 2017; Hérault and Ribar, 2017; Meyer, Wyse and Logani, 2023; Moschion and Johnson, 2019). For example, Meyer, Wyse and Logani (2023) use administrative data to document markedly elevated mortality among those experiencing

¹These findings are in line with event studies of the labor market trajectories of individuals becoming homeless after their landlord is taking back their flat. The event studies show that homelessness produces sharp and persistent declines in employment and earnings in the years following the shock. Labor market trajectories do not recover, providing additional evidence that homelessness leads to prolonged spells of unemployment.

homelessness. Because quasi-experimental designs are rarely feasible with existing data, this evidence is largely correlational. Eviction studies provide causal evidence, but an eviction order affects labor market outcomes through several channels, of which homelessness is only one, so they cannot isolate its role. Exploiting quasi-random variation in whether a first request for shelter is met, I show that homelessness raises job separations and lowers job finding rates.

Second, this paper contributes to the literature evaluating housing interventions for unhoused individuals ([Brounstein and Wieselthier, 2024](#); [Christopher, Duggan and Martin, 2025](#); [Cohen, 2024](#); [Costantini, 2026](#); [Faraji, Ridgeway and Wu, 2018](#)). Most of this work evaluates Housing First programs, which provide permanent housing, and finds improvements in housing stability and reductions in service use. Closest to this paper, [Christopher et al. \(2025\)](#) study the effects of temporary shelter on crime and health. I estimate the effect of temporary shelter on labor market outcomes, which has not previously been examined.

Third, I contribute to the measurement of homelessness. Identifying unhoused individuals is itself a central obstacle in this literature ([O’Flaherty, 2019](#)), and recent work has made progress by exploiting administrative records ([Christopher et al., 2025](#); [Costantini, 2026](#); [Meyer, Wyse, Grunwaldt, Meyer and Wu, 2023](#); [Von Wachter et al., 2020](#)). Two limitations remain. These records typically identify individuals at a single point in time, and some oversample the chronically homeless, missing those who are homeless only briefly. In addition, records of shelter stays are silent on the unsheltered. The data used here make advances on both fronts: because they cover the universe of calls to the national emergency shelter hotline and all shelter stays, they capture transitional as well as chronic spells, and they identify individuals whose request was denied and who therefore remained more exposed to street homelessness. Merging these records with administrative employment and unemployment data is what makes the quasi-experimental designs in this paper feasible.

Finally, I contribute to the literature on the consequences of job loss. A large body of work documents its lasting costs in earnings, health, and mortality ([Bennett and Ouazad, 2020](#); [Bhalotra, Britto, Pinotti and Sampaio, 2025](#); [Black, Devereux and Salvanes, 2015](#); [Britto, Pinotti and Sampaio, 2022](#); [Browning, Moller Dano and Heinesen, 2006](#); [Jacobson, LaLonde and Sullivan, 1993](#); [Roulet, 2020](#)). I identify homelessness as one of its most severe and least documented consequences. A related literature examines the links between housing and labor markets for low-income households ([Dauth, Mense and Wrede, 2024](#); [Dustmann, Fitzenberger and Zimmermann, 2022](#)). To interpret the two sets of estimates jointly, I extend a [McCall \(1970\)](#) job search

model to account for the housing situation of the job seeker, and use it to quantify how much the feedback between homelessness and unemployment prolongs unemployment spells.

2 Institutional Background

With one of the highest rates of homelessness per capita and the majority being sheltered, France presents an interesting setting to study homelessness tracking entries and exits through its shelter system.

2.1 Homelessness in France

When estimating the size of the homeless population, a central challenge is the underlying measurement and definition of homelessness. France adopts the European ETHOS framework ([Feantsa, 2005](#)), which disaggregates homelessness into several distinct categories. The first category consists of individuals 'living in the streets or public spaces without a shelter that can be defined as living quarters.' The second category includes those in emergency accommodation, 'people with no place of usual residence who move frequently between various types of accommodation.' The third category encompasses individuals in accommodation for the homeless, where the 'period of stay is time-limited, and no long-term housing is provided.' Finally, the framework includes three additional categories that are not always included in narrower definitions of homelessness: individuals living in institutions, those in non-conventional dwellings, and people living temporarily with family or friends due to a lack of housing.

In the French context, an individual is defined as homeless if they fall within the first three ETHOS categories or reside in a non-conventional dwelling due to a lack of housing. Based on 2022 estimates, approximately 5,500 individuals were classified under Category 1, while roughly 316,000 individuals were situated in Categories 2 and 3. Within this latter group, approximately 113,000 were housed in temporary accommodation specifically for asylum seekers. Finally, an estimated 11,200 individuals lived in non-conventional dwellings — such as caravans, cars, or tents — due to a lack of permanent housing ([Dihal, 2022](#)).

The French homeless population has expanded substantially over the last decade, with the total number of individuals more than doubling from 143,000 in 2012 to approximately 350,000 in 2025. Of these 350,000 individuals, roughly 334,000 reside in shelters. This figure excludes an estimated 590,000 individuals who are staying with friends or family due to housing difficulties

([Fondation pour le logement \(2025\)](#)).

2.2 Shelter Allocation System in France

In France, homeless shelter requests are tracked and managed by the SIAO (*Service Intégré d'Accueil et d'Orientation*; in English: *the Integrated Service for Reception and Orientation*). The system comprises 96 local units, organized at the departmental level—the primary administrative unit in France. These units maintain a comprehensive inventory of all available shelter capacity, spanning both short-term emergency facilities and time-limited transitional housing. Based on individual requests, the SIAO then coordinates the placement and orientation of individuals into these available accommodations. [Figure 1](#) summarizes the pathways available to an individual following a housing shock.

Individual entry into the shelter system typically begins with a request via the national intake hotline, *115*, which directs individuals toward emergency accommodation. The *115* shelter intake hotline is the fourth most used emergency hotline in France, after the ambulance, police, and fire lines, and is staffed by roughly 950 call-center agents. The hotline operates around the clock since its creation in 1997, but individual calls have only been recorded systematically since 2020. The scale of unmet demand is large: on an average day, 60% of requests are not met.

The vast majority of individuals is first observed through calling the *115* hotline. However, alternatively social workers, often based in day centers that serve unsheltered populations, may initiate requests for emergency or longer-term transitional housing. Similarly, social workers within the emergency shelter system can also open applications for longer-term placement in transitional housing facilities.

In total, as of year-end 2024, the French shelter system comprised 203,798 places. This capacity is divided between emergency accommodation, consisting of emergency shelters (25%) and social hotels (42%), and longer-term, transitional housing (32%). However, the system remains highly saturated, with approximately 7,000 requests for emergency shelter going unmet on any given night. This figure likely represents a lower bound, as many individuals experiencing homelessness report that they have either never called or have stopped calling the *115* hotline after repeated failed attempts ([Fondation pour le logement \(2025\)](#)).

3 Data and Descriptive Statistics

I construct a novel dataset that merges, for the first time, nationwide homeless shelter records with administrative employment and unemployment data. This linkage enables a detailed reconstruction of the labor market trajectories of individuals experiencing homelessness, highlighting an interaction between precarious labor market conditions and an increased risk of homelessness.

3.1 Data

This project constructs a novel dataset that merges, for the first time, nationwide homeless shelter records with administrative employment and unemployment data.

3.1.1 Nationwide Homeless Management Information System

The nationwide homeless management information system records every shelter request made by an individual, regardless of the outcome of the request. Thanks to a unique individual and household identifier, this dataset therefore allows us to track an individual's entries into and exits from homelessness over time.

Through an agreement with *Dihal*, I have access to the pseudonymized homeless management information system (SI-SIAO). The data has two main components. The 115 call data are structured by request, with a unique household and individual identifier; each record includes the date and time of the call, the type of request and its status (met or not met), and background information including the stated reason for the request, the individual's housing situation at the time of the call, and household composition. The shelter request data provide a parallel set of records covering emergency-shelter and transitional-housing requests; these records contain the same fields as the 115 data, plus additional information on the eventual shelter stay where placement occurred (shelter name, address, date of entry and exit, and the individual's situation at exit). Each individual is assigned a unique identifier, which enables me to track shelter requests, entries, and exits over time, and to measure homelessness both at a single point in time and as a flow.

The administrative shelter data provide near-complete coverage of *sheltered* homelessness (ETHOS categories 2 and 3), which constitutes the vast majority of homelessness in France. They also provide partial coverage of *unsheltered* homelessness (ETHOS categories 1 and 5) for

individuals who have made at least one shelter request, even if the request was not met. A primary measurement challenge is that individuals are only observable in the data if they actively request shelter or reside within a facility during a given period; consequently, I cannot identify individuals sleeping in tents or on the street who never contact the 115 hotline. In terms of sample selection, however, this is less of a concern in the French context: the vast majority of the homeless population in France is sheltered.

This fact makes France a particularly compelling case for studying homelessness through administrative homelessness management systems, especially when compared to countries like the United States, which exhibit significantly higher rates of unsheltered homelessness. To address potential under-reporting, I leverage the seasonal variation in winter shelter programs, which see an expansion of capacity and an influx of requests from the unsheltered population. Finally, I use the COVID-19 pandemic as a robustness check; during this period, France significantly expanded shelter capacity through the large-scale requisitioning of hotels as part of a policy effort to eliminate unsheltered homelessness.

A further data challenge concerns the reliable measurement of *exits* from homelessness, particularly out of emergency shelter: individuals may stop calling 115 either because they have found housing or because they have given up after repeated unmet requests, and these alternatives cannot be distinguished. Throughout the empirical analysis I therefore focus on entry into homelessness, which is reliably captured.

Figure 2 plots the volume of 115 requests over time, decomposed by household type. Single men constitute the largest share of callers throughout the period, and the overall volume of requests has reached almost 200,000 requests per months in 2024. Figure 3 compares the count of unique individuals appearing in the SI-SIAO system per year against the aggregate Dihal estimate of the homeless population: by 2022–2023 the SI-SIAO records cover essentially the entire estimated homeless population, confirming that for the purposes of this paper the data covers the vast majority of individuals experiencing homelessness in France.

3.1.2 Administrative Employment and Unemployment Records

While the homelessness management information system provides a detailed view of entries into and exits from the shelter system, it lacks information on individuals' labor market outcomes. To reconstruct these trajectories, the shelter records are merged with administrative employment and unemployment data within a highly secure environment.

Individuals are linked to the employment and unemployment records if they held a work contract or were registered with the French unemployment agency at any point since 2017.

To analyze labor market outcomes, I utilize the *Déclaration Sociale Nominative* (DSN) for the 2017–2025 period. The DSN consists of monthly employer-submitted declarations for every employee, enabling me to track the universe of work contracts for the sampled individuals. These data allow me to assess not only monthly employment status but also specific contract conditions, such as the use of short-term precarious contracts and precise earnings levels. I furthermore have detailed information on individual’s unemployment spells from the French Public Employment Services Registry, including information on the assistance and benefits they received.

3.1.3 Record-Linkage Procedure

In this paper, I merge the largest existing sample of individuals experiencing homelessness with administrative employment and unemployment records from France for the period 2017 to 2025. By linking records at the individual level based on names and date of birth in a secure environment, I can not only study detailed labor market trajectories around the first homelessness episode, but also consider novel identification strategies for understanding the existence of a homelessness trap.

As the *SI-SIAO* does not collect social security identifiers compatible with administrative systems, the linkage proceeds in three steps, ordered from most to least restrictive. Step 1 performs an exact match on first name, last name, and date of birth, recovering approximately 94% of all eventual matches. Step 2 applies the Jaro–Winkler similarity metric to first and last name to identify strongly similar (but not identical) names, combined with an exact match on date of birth; this step recovers approximately 5% of matches. Step 3 uses the Jaro–Winkler similarity for names combined with an exact match on either month or year of birth and an exact match on phone number; this step recovers approximately 0.08% of matches. Of the 3,262,677 unique individuals appearing in the *SI-SIAO* since 2014,² 1,111,354 are successfully linked to administrative records.

Before reconstructing individual’s labor market trajectories, I first assess whether the matching process introduces selection. After excluding children and recent migrants — who have a lower probability of appearing in French administrative labor records — [Table 1](#) compares the

²While the system was established in 2014, it was not universally adopted until 2020. Since then, shelter allocation has been strictly required to occur through this centralized platform.

characteristics of the matched and unmatched samples. The results indicate that both groups are balanced across key demographics, including household composition, age, and gender. The only significant divergence is that non-French nationals are less likely to be matched. This discrepancy is likely attributable to the imperfect identification of all recent arrivals; consequently, the unmatched group may still contain individuals who lack the legal right to work, precluding them from formal employment and unemployment agency registration.

3.1.4 Descriptive Statistics

The data merge allows me to describe labor market trajectories of individuals experiencing homelessness in detail, highlighting the interaction between precarious labor market conditions and an increased risk of homelessness.

[Figure 4](#) decomposes monthly 115 requests by the caller's labor-market attachment. Two facts stand out. First, a substantial fraction of individuals entering the shelter system are not detached from the formal labor market at the moment of their housing crisis: in 2024, 8.6% were employed at the time of their request, and 26.7% report a work history in the prior two years. Second, the number of requests by individuals with a recent work history has increased in absolute numbers over time.

Among matched individuals with prior labor-market history, average monthly earnings prior to the first shelter request stand at €875, with a median of €777. These low earnings levels are primarily driven by the prevalence of very short-term contracts; the median contract duration in the sample is only six days. Such short durations reflect a structural feature of the French 'dual' labor market, which is characterized by a sharp divide between stable open-ended contracts (CDI) and fixed-term contracts (CDD) that are often extremely brief.³ Indeed, as [Figure A1](#) illustrates, individuals on fixed-term contracts face a significantly higher risk of entering homelessness in the subsequent year compared to those on open-ended contracts, a disparity that persists across the entire income distribution.

Lastly, [Figure 5](#) investigates the labor market trajectories of individuals following their first observed spell of homelessness. For this analysis, the onset of homelessness is proxied by the date of the individual's first recorded request for shelter. To identify potential shifts in labor market outcomes, I restrict the sample to individuals with at least a marginal labor market attachment,

³[Bergeaud, Cahuc, Malgouyres, Signorelli and Zuber \(2024\)](#) report a median contract length of only three days for fixed-term contracts in France, which is consistent with the patterns observed in my sample.

defined as having worked for at least 180 days within the preceding five years. The figures reveal a sharp and substantial decline in both employment and earnings around the time of the first shelter request. These outcomes remain depressed several years after the initial request, suggesting individuals being trapped in unemployment and homelessness. While baseline earnings are notably low, it is important to note that these figures represent unconditional earnings; in the two years prior to the first shelter request, only approximately 45% of this marginally attached group were actively employed. This descriptive pattern is suggestive but potentially consistent with reverse causation or anticipatory deterioration — motivating the formal identification strategies developed in the next sections.

4 Conceptual Framework: A Search Model Where Housing Matters

To guide and interpret the main empirical analysis identifying the vicious cycle between homelessness and unemployment, I extend a [McCall \(1970\)](#) job search model to incorporate the housing situation of a worker.⁴ The model shows how feedback loops between the labor and housing market prolongs both unemployment and homelessness.

4.1 Environment

Workers and states. The economy is populated by a continuum of infinitely lived, risk-neutral workers who discount the future with factor $\beta \in (0, 1)$. A worker's employment status is $e \in \{E, U\}$ (employed or unemployed) and her housing status is $h \in \{H, NH\}$ (housed or unhoused). This gives four states: (E, H) , (U, H) , (E, NH) and (U, NH) . Employed workers earn a monthly wage w , and housed workers pay a rent r . Throughout, subscripts H and NH refer to housing status.

Labor market. Unemployed workers receive benefits b . Each month, a housed, unemployed worker receives a job offer with probability λ and an unhoused, unemployed worker with probability $\kappa\lambda$. The parameter $\kappa \in [0, 1]$ scales down the offer probability of unhoused workers: $\kappa = 1$ means no penalty and $\kappa = 0$ means no offers. Monthly wage offers are drawn from a distribution F , which is log-normal and truncated below at the benefit level b . Each month, an employed worker loses her job with probability s_H if housed and s_{NH} if unhoused. There is no on-the-job

⁴I use a partial-equilibrium search model rather than an equilibrium matching model as the shocks I study are individual and unlikely to move market wages and rents.

search.

Unhoused workers may receive fewer job offers and lose jobs more often: without a stable address, reliable contact or adequate rest, searching for and holding a job becomes harder. Qualitative studies document such barriers, including among homeless people who work (Goldstone, 2025; Poremski, Whitley and Latimer, 2014; Shier, Jones and Graham, 2012; Snow and Anderson, 1993). Housing loss also predicts subsequent job loss (Desmond and Gershenson, 2016) and lower earnings (Collinson et al., 2024). None of these studies, however, isolates the causal effect of homelessness on job finding and job separation, which remain thus an open empirical question. The model therefore allows for $\kappa < 1$ and $s_{NH} > s_H$ without imposing them. I refer to the benchmark $\kappa = 1$ and $s_{NH} = s_H$, in which housing status does not affect the labor market, as *housing neutrality*.

Housing market. Each month, a housed worker experiences a housing shock with probability θ , such as a landlord selling the flat, a rent increase or a household separation, and draws a new rent from a distribution G .⁵ An unhoused worker pays no rent, incurs a monthly disutility φ_{NH} , and receives a housing offer with probability π , with rent drawn from G . Rents are zero with probability p_0 , representing workers housed at no cost in a shelter or with family or friends, and otherwise log-normally distributed:

$$G(r) = p_0 + (1 - p_0) \Lambda(r; \mu_G, \sigma_G), \quad r \geq 0,$$

where $\Lambda(\cdot; \mu_G, \sigma_G)$ is the log-normal distribution function. Since the log-normal has positive density on $(0, \infty)$, G is strictly increasing on $(0, \infty)$. Wage offers and rent draws are independent of each other and independently distributed across workers and over time. A dwelling is affordable if its rent does not exceed income, so $G(y)$ is the probability that a rent draw is affordable for a worker with income y . The monthly probabilities of losing and of regaining housing at income y are therefore

$$\delta(y) \equiv \theta [1 - G(y)], \quad g(y) \equiv \pi G(y). \quad (1)$$

Since G is strictly increasing, a higher income makes more dwellings affordable: $\delta(y)$ strictly

⁵The model thus captures two channels of entry into homelessness: income losses, such as job loss or benefit exhaustion, which operate through the affordability constraint, and housing shocks, such as a separation or divorce, which operate through θ . For evictions, these are the two main reasons documented by ?. Other entry channels, such as severe mental health shocks, matter mainly in highly disadvantaged populations with weak labor market attachment (see, e.g., ?, for a study of this population). Since the model focuses on the interaction between the labor and housing markets, it abstracts from these channels, which are not the main channels of entry for the population of interest: workers who are employed or were recently employed.

decreases and $g(y)$ strictly increases in income.

Assumption 1 (Hand-to-mouth). *Workers consume their income net of rent and can neither save nor borrow.*

Assumption 2 (Earnings exceed benefits). *Every job offer pays monthly earnings above unemployment benefits: $F(b) = 0$.*

Assumption 3 (Large disutility of homelessness). *The disutility of homelessness φ_{NH} is large enough that every worker's reservation rent exceeds her income: workers accept and keep every dwelling they can afford.*

Assumption 1 implies that affordability binds whenever income changes. Assumption 2 implies that employed workers can afford strictly more dwellings than unemployed workers. It does not require jobs to be full time: part-time and short jobs are included as long as their monthly earnings exceed benefits. Assumption 3 makes the reservation rent a corner solution at income, so housing transitions are governed by affordability alone. I verify it at the estimated parameters; Appendix ?? defines the reservation rent and treats the interior case.

4.2 Timing and decisions

At the start of month t , a worker is described by her state (e, h) , her wage w if employed and her rent r if housed. Each month unfolds in three steps.

1. *Payoffs.* During month t , the worker earns her income, w or b , pays her rent r if housed or incurs the disutility φ_{NH} if unhoused, and consumes the remainder.
2. *Labor stage.* An employed worker loses her job with probability s_H if housed or s_{NH} if unhoused. An unemployed worker receives a job offer with probability λ if housed or $\kappa\lambda$ if unhoused, observes the wage w' and decides whether to accept it.
3. *Housing stage.* A housed worker experiences a housing shock with probability θ and an unhoused worker receives a housing offer with probability π . The worker then lives in a dwelling in month $t + 1$ if and only if its rent does not exceed her income in month $t + 1$, which is determined by the labor stage.⁶

⁶This timing is a simplification. A worker who can no longer pay her rent loses her dwelling within the month, whereas in practice rent arrears accumulate and eviction procedures are long and costly, so the model describes the fastest possible response to an income loss. Many housing shocks are, however, immediate, such as a household separation or being asked to leave by family or friends. The job-loss design traces the actual timing by following workers over several months after job loss.

All shocks are independent, so several events can occur in the same month: a worker can, for instance, lose her job and her dwelling in the same month. The model delivers the probability of each state at the start of month $t + 1$, which is what the monthly data record.

Under Assumption 3, the only decision is whether to accept a job offer, which reduces to choosing a reservation wage. It is taken in the labor stage, before the housing shock or offer is realized, so the worker compares expected values over the housing stage. Housing transitions involve no decision: the worker lives in every dwelling she can afford. Without Assumption 3, workers would also decide whether to live in an affordable dwelling, which allows for voluntary homelessness; Appendix ?? treats this case.

Figure 6 summarizes the resulting monthly transitions between the four states.

4.3 Value functions and reservation wages

Let $W_H(w, r)$ denote the value of an employed, housed worker with wage w and rent $r \leq w$; $U_H(r)$ the value of an unemployed, housed worker with rent $r \leq b$; $W_{NH}(w)$ the value of an employed, unhoused worker with wage w ; and U_{NH} the value of an unemployed, unhoused worker.

Affordability. Let $\mathcal{U}(r)$ denote the value of an unemployed worker who starts the next month with a dwelling at rent r , and $\mathcal{W}(w, r)$ the corresponding value of an employed worker with wage w . By the affordability rule, the worker is housed next month if and only if she can pay the rent:

$$\mathcal{U}(r) = \begin{cases} U_H(r) & \text{if } r \leq b, \\ U_{NH} & \text{if } r > b, \end{cases} \quad \mathcal{W}(w, r) = \begin{cases} W_H(w, r) & \text{if } r \leq w, \\ W_{NH}(w) & \text{if } r > w. \end{cases} \quad (2)$$

Bellman equations. Let $\mathbb{E}_G$ denote the expectation over a new rent r' and $\mathbb{E}_F$ the expectation

over a wage offer w' . The value functions satisfy

$$\begin{aligned} W_H(w, r) = & w - r + \beta(1 - s_H) [(1 - \theta) W_H(w, r) + \theta \mathbb{E}_G \mathcal{W}(w, r')] \\ & + \beta s_H [(1 - \theta) U_H(r) + \theta \mathbb{E}_G \mathcal{U}(r')], \end{aligned} \quad (3)$$

$$\begin{aligned} U_H(r) = & b - r + \beta \lambda \mathbb{E}_F \max \left\{ (1 - \theta) W_H(w', r) + \theta \mathbb{E}_G \mathcal{W}(w', r'), \right. \\ & \left. (1 - \theta) U_H(r) + \theta \mathbb{E}_G \mathcal{U}(r') \right\} \\ & + \beta(1 - \lambda) [(1 - \theta) U_H(r) + \theta \mathbb{E}_G \mathcal{U}(r')], \end{aligned} \quad (4)$$

$$\begin{aligned} W_{NH}(w) = & w - \varphi_{NH} + \beta(1 - s_{NH}) [(1 - \pi) W_{NH}(w) + \pi \mathbb{E}_G \mathcal{W}(w, r')] \\ & + \beta s_{NH} [(1 - \pi) U_{NH} + \pi \mathbb{E}_G \mathcal{U}(r')], \end{aligned} \quad (5)$$

$$\begin{aligned} U_{NH} = & b - \varphi_{NH} + \beta \kappa \lambda \mathbb{E}_F \max \left\{ (1 - \pi) W_{NH}(w') + \pi \mathbb{E}_G \mathcal{W}(w', r'), \right. \\ & \left. (1 - \pi) U_{NH} + \pi \mathbb{E}_G \mathcal{U}(r') \right\} \\ & + \beta(1 - \kappa \lambda) [(1 - \pi) U_{NH} + \pi \mathbb{E}_G \mathcal{U}(r')]. \end{aligned} \quad (6)$$

Each equation consists of the flow payoff and the discounted value after the two stages. The labor stage determines the weights: the job-separation probability s_H or s_{NH} for employed workers, and the offer probability λ or $\kappa \lambda$ for unemployed workers. Each term in square brackets is the expected value after the housing stage: a housed worker keeps her dwelling with probability $1 - \theta$, provided she can still afford it, and otherwise draws a new rent r' ; an unhoused worker receives a housing offer with probability π . Affordability enters through (2).

Equation (3) contains the channel from job loss to homelessness: a housed worker who loses her job enters the housing stage with income b , so she loses her dwelling if $r > b$, and any dwelling she draws later must be affordable at b . Equations (5) and (6) contain the reverse channel: an unhoused worker loses her job with probability s_{NH} and receives offers with probability $\kappa \lambda$.

Lemma 1 (Reservation wages). *An unemployed worker accepts a wage offer w' if and only if $w' \geq w_H^*$ when housed and $w' \geq w_{NH}^*$ when unhoused. The reservation wage w_H^* does not depend on the worker's rent.*

The housed reservation wage does not depend on the rent, because a worker with $r \leq b$ pays the same rent whether or not she accepts the offer (Lemma ?? in Appendix ?? shows that the values

of housed workers are linear in the rent). Appendix ?? characterizes both reservation wages. Relative to a housed worker, three forces shape the reservation wage of an unhoused worker. Fewer offers ($\kappa < 1$) and a higher separation probability ($s_{NH} > s_H$) reduce the value of continued search, which lowers w_{NH}^* . Employment also has housing value: it raises the probability of rehousing from $g(b)$ to $g(w)$ for unhoused workers, and it insures housed workers against losing their home after a housing shock. The resulting ranking of w_{NH}^* and w_H^* is a quantitative question. The monthly job-finding probabilities are

$$f_H \equiv \lambda [1 - F(w_H^*)], \quad f_{NH} \equiv \kappa \lambda [1 - F(w_{NH}^*)]. \quad (7)$$

Extension: benefit exhaustion. Unemployment insurance is paid for a limited duration. In an extension, unemployed workers receive b_1 during the first T months of unemployment and $b_0 < b_1$ afterwards. The values of unemployed workers then depend on elapsed unemployment duration, and Assumption 2 becomes $F(b_1) = 0$ (Appendix ??). The base model corresponds to $T \rightarrow \infty$ with $b = b_1$.

4.4 Steady state

Wages and rents are drawn from exogenous distributions, so the model is solved in partial equilibrium.

Definition 1 (Steady state). *A partial-equilibrium steady state consists of value functions W_H , U_H , W_{NH} and U_{NH} , reservation wages w_H^* and w_{NH}^* , and a distribution Ψ of workers over states, wages and rents such that (i) the value functions solve (3)–(6); (ii) the reservation wages are optimal given the value functions; and (iii) Ψ is invariant under the monthly transition probabilities implied by the shocks, the affordability rule and the reservation wages.*

Appendix ?? states the monthly transition probabilities. On a discretized state space, the steady state exists and is unique (Appendix ??). Let $m(e, h)$ denote the steady-state population share in state (e, h) . Two averages over Ψ are used below: $\bar{\delta}$, the average monthly probability that an employed, housed worker loses housing, and ρ , the share of employed, housed workers whose rent exceeds b .

4.5 The vicious cycle

The vicious cycle arises when both directions operate: income losses push workers into homelessness, and homelessness keeps them out of work. To show how the two directions reinforce each other, I characterize how long workers remain in each state. For an unemployed worker with housing status h , let the *expected remaining unemployment duration* ℓ_h denote the expected number of months, including the current one, until she finds a job. For an unemployed, unhoused worker, let the *expected duration of homelessness* τ denote the expected number of months, including the current one, until she is housed; she may find and lose jobs in the meantime. Both follow from the monthly transition probabilities in Appendix ???. The following proposition characterizes them in the base model.

Proposition 1 (Vicious cycle).

- (i) Homelessness prolongs unemployment. *The expected remaining unemployment durations of unhoused and housed workers differ by*

$$\ell_{NH} - \ell_H = \frac{1/f_{NH} - 1/f_H}{1 + (1 - f_{NH})g(b)/f_{NH} + (1 - f_H)\delta(b)/f_H},$$

so $\ell_{NH} > \ell_H$ if and only if $f_{NH} < f_H$.

- (ii) Unemployment prolongs homelessness, and the feedback deepens it. *Holding the distribution of accepted wages fixed, the expected duration of homelessness τ decreases in f_{NH} . If all accepted wages imply the same rehousing probability, τ also increases in s_{NH} .*

Part (ii) relies on employment raising the rehousing probability, $g(w) > g(b)$ for every accepted wage, which follows from Assumption 2. The fewer jobs unhoused workers find and the more often they lose them, the longer they remain unhoused. By part (i), the longer they remain unhoused, the longer they remain unemployed if homelessness lowers job finding. Appendix ??? gives τ in closed form.

??? uses the estimated model to quantify how much the vicious cycle contributes to unemployment and homelessness.

4.6 Discussion of the modelling choices

Hand-to-mouth workers. The population studied has low incomes that consist mostly of earnings or benefits. As wealth is well predicted by earnings, it is therefore reasonable to assume that the liquid wealth of the studied population is rather small. Therefore, I abstract from wealth and savings in this model. An income loss therefore translates quickly into rent arrears, which Assumption 1 captures. Savings would delay housing loss after job loss; the model describes the fastest possible response, while the empirical estimates capture the actual timing.

Housing status. The model distinguishes only between housed and unhoused workers. Shelters enter the model as housing at zero rent, thus a mass point of G at zero, and access to it is rationed through π .⁷ Treating homelessness as a single state abstracts from differences between forms of precarious housing.

Benefits. Unemployment insurance in France is proportional to the previous wage, whereas the model uses a single benefit level b . For the low-wage population studied, benefits are close to a common level. Assumption 2 abstracts from jobs whose monthly earnings fall below benefits, such as jobs with very few hours, as these are typically combined with unemployment benefits. Short contracts that end quickly are captured by the separation probabilities s_H and s_{NH} . The extension with benefit exhaustion captures the main source of variation in benefits over an unemployment spell.

Exogenous wages, rents and separations. Wages and rents are drawn from fixed distributions because the shocks studied are individual, and separations are exogenous because the empirical analysis measures separation probabilities rather than their causes. The rehousing probability π is common to all workers, although social housing in France is allocated partly through priority rules, and the housing-shock probability θ may be higher for workers facing health or family shocks. Both could vary by worker type without changing the structure of the model.

4.7 Testable implications

The model yields three predictions. The first two concern the direction from income losses to homelessness. The third concerns the vicious cycle, which requires the reverse direction as well:

⁷Alternatively, shelters could be modeled as a separate state with their own offer and separation probabilities, between housing and rough sleeping. The two-state version is sufficient for the shelter-access design, which compares workers with and without a shelter place.

homelessness must worsen labor market outcomes. Proofs are in Appendix ??.

Only one feature of the rent matters for housing transitions: whether it exceeds benefits, which determines whether job loss leads to losing the dwelling.

Prediction 1 (Job loss causes homelessness). *Consider a housed, employed worker with wage w and rent r . Relative to keeping her job, losing it*

- (a) *raises her probability of being unhoused in the following month from $\delta(w)$ to $(1 - \theta) \mathbf{1}\{r > b\} + \delta(b)$; on average across employed, housed workers, from $\bar{\delta}$ to $(1 - \theta) \rho + \delta(b)$;*
- (b) *raises her monthly probability of losing housing in each further month of unemployment from $\delta(w)$ to $\delta(b)$.*

Prediction 2 (Benefit exhaustion). *In the extension with limited benefit duration, consider an unemployed worker who does not find a job. At benefit exhaustion,*

- (a) *if housed with rent r , her probability of becoming unhoused in the following month rises from $\delta(b_1)$ to $(1 - \theta) \mathbf{1}\{r > b_0\} + \delta(b_0)$, and remains at $\delta(b_0)$ in later months;*
- (b) *if unhoused, her monthly probability of being rehoused falls from $g(b_1)$ to $g(b_0)$.*

Prediction 3 (Vicious cycle). *If homelessness feeds back into the labor market, the labor market outcomes of unhoused workers differ from those of housed workers on two margins:*

- (a) *job loss: unhoused workers lose their jobs more often if $s_{NH} > s_H$;*
- (b) *job finding: the effect of homelessness on job finding is ambiguous. Fewer offers ($\kappa < 1$) lower job finding, whereas a lower reservation wage raises the share of offers that unhoused workers accept. Unhoused workers find jobs less often if the offer channel dominates:*

$$\kappa < \frac{1 - F(w_H^*)}{1 - F(w_{NH}^*)}. \quad (8)$$

By Proposition 1, more frequent job loss lengthens homelessness, and less frequent job finding lengthens both unemployment and homelessness.

The remainder of the paper takes these implications to the data. ?? tests Predictions 1 and 2 using job losses, and Section 6 tests Prediction 3 using quasi-random access to shelters. ?? combines both sets of estimates to estimate the model and quantify the contribution of the vicious cycle.

5 The Effect of Job Loss on Homelessness

To recover the casual effect of job loss on homelessness risk, I restrict the analysis to workers that were laid off for an economic reason, that is a reason independent of their individual performance, and compare them to similar workers working in firms not experiencing economic layoffs. I find that job loss increases the risk of becoming homeless by 50% in the three subsequent years after the layoff.

5.1 Research Design

To estimate the causal effect of job loss on homelessness, I focus on workers that were laid off during economic layoffs happening in their firm. Using a one-to-one matching approach, I then compare the homelessness risk of those laid-off workers to workers that are similar on a wide range of socio-demographic and job characteristics at the time of the layoff.

The ideal experiment to estimate the causal effect of job loss on homelessness would involve the random displacement of workers. In practice, however, job loss is rarely a random event; confounding factors — such as health or family shocks — may simultaneously influence both the probability of displacement and the risk of homelessness. To address this endogeneity, I exploit layoffs for economic reasons to isolate the impact of job loss from other idiosyncratic shocks. This identification strategy relies on the assumption that workers displaced in an economic layoff are terminated solely for economic reasons unrelated to their individual performance. Specifically, the counterfactual assumes that, in the absence of the layoff, these individuals would have remained at their firms. This assumption of parallel trends can be evaluated visually by comparing pre-event trajectories across the outcomes of interest.

The primary specification uses the universe of economic layoffs between 2017 and 2023 where at least two workers were dismissed from a given firm in a given month — the legal threshold for an economic layoff under French law.⁸ I restrict the sample to workers who were 20 to 60 years old at the time of displacement, held an open-ended contract and who were French. I specifically restrict the sample to French workers due to a better match quality for this group with the administrative employment records. Each treated worker is matched one-

⁸The full set of legal grounds for economic layoff that I include is: *licenciement suite à liquidation judiciaire ou à redressement judiciaire; licenciement suite à fermeture définitive de l'établissement; licenciement pour motif économique; licenciement pour fin de chantier ou d'opération; autre fin de contrat pour motif économique; rupture pour motif économique dans le cadre d'un contrat de sécurisation professionnelle (CSP); and rupture anticipée d'un contrat de travail ou d'un contrat de mission suite à fermeture de l'établissement.*

to-one to an observably similar control worker who was not displaced in the same calendar year and was not employed in a firm engaged in economic layoffs during the analysis period. Matching is performed via exact matching on earnings bins, four-digit occupation, geographic area (*département*), age bins, job-tenure bins, and high-school completion. In cases of multiple potential matches, one is selected at random. This procedure yields a matched sample of 53,619 treated workers (drawn from a pool of 240,370 displaced workers) and an equal number of controls. As noted by [Britto et al. \(2022\)](#), this design is robust to recent methodological concerns regarding staggered difference-in-differences. By utilizing a group of never-treated workers, this approach avoids the biases associated with heterogeneous treatment effects in standard two-way fixed effects models ([Callaway and Sant’Anna, 2021](#); [De Chaisemartin and d’Haultfoeuille, 2020](#)).

To estimate dynamic treatment effects, I run the following regression:

$$Y_{it} = \alpha + \gamma Treat_i + \sum_{t=-P, t \neq 0}^T \delta_t (Treat_i \times Time_t) + \phi_t + \epsilon_{it} \quad (9)$$

Y_{it} denotes the outcome of interest of a worker i at quarter t , relative to the quarter of the economic layoff event. Each matched control worker is assigned a ‘placebo’ displacement date corresponding to the actual layoff date of their treated counterpart. $Treat_i$ is a binary indicator equal to one for individuals displaced in an economic layoff and zero for matched controls. I further include year fixed effects, ϕ_t , to account for macroeconomic shocks specific to the layoff cohort. Standard errors are clustered at the individual level.

To get an understanding of the overall magnitude of the effect, I furthermore estimate the following pooled equation:

$$Y_{it} = \alpha + \gamma Treat_i + \beta (Treat_i \times Post_t) + \gamma Post_t + \epsilon_{it} \quad (10)$$

The primary modification to specification 9 is the aggregation of the post-event periods into a single estimate. Specifically, I replace the individual time-to-event indicators with a $Post_t$ dummy, which takes a value of one for all years following the economic layoff.

To evaluate the robustness of this design, I employ a second identification strategy that leverages the quasi-random timing of displacement events following the framework of [Callaway and Sant’Anna \(2021\)](#) and [Sant’Anna and Zhao \(2020\)](#), comparing ‘not-yet-treated’ units to those already treated. This approach utilizes multiple years of displacement events without imposing

matching restrictions, and serves both as a robustness check on the matched design and as a way of testing the underlying counterfactual that displaced individuals would have remained at their firm in the absence of the layoff. A significant advantage of this approach is that, unlike the matching design, pre-trends are not mechanically aligned by the matching algorithm, providing a more transparent test.

5.2 Main Results

I find a 50% increase in the probability of becoming homeless within the three years following the economic layoff event. While the baseline probability of becoming homeless is very low, this finding highlights that homelessness is an important factor to consider when assessing the negative consequences of job loss.

Table 2 reports the pooled estimates from Equation 10. Job displacement raises the probability of entering homelessness by approximately 50% relative to the baseline mean of 0.05% — a striking magnitude in a country with one of the world’s most generous social safety nets. The labor-market effects are also substantial: employment falls by 32%, employment in stable open-ended contracts falls by 41%, and average monthly earnings fall by 31%.

Figure 7 traces the dynamic path of labor-market outcomes around the economic layoff event. Consistent with the existing literature on job displacement, all three outcomes drop sharply in the first quarter following the layoff and recover only partially over the subsequent three years. Pre-event trends are flat, supporting the parallel-trends assumption.

Figure 8 presents the dynamic effect on homelessness risk. The effect on homelessness emerges within the first year and grows steadily: by approximately three years after the displacement event, homelessness probability has roughly *doubled* relative to the baseline. Although the point estimates are noisy in individual quarters — reflecting the very low baseline incidence (0.05%) — the upward trajectory is consistent and the pattern is qualitatively similar when restricting to male workers (right panel), though the baseline is somewhat higher (0.07% vs. 0.05%). Restricting attention to the *first* call for shelter (Figure 9), the increase in homelessness risk is concentrated one to two years after the layoff, consistent with the exhaustion of unemployment insurance benefits. Robustness checks excluding the COVID-19 years and using only treated units can be found in Appendix B and yield the same qualitative conclusions.

5.3 Heterogeneity

Heterogeneity analysis shows that the effects are driven by individuals who have low income and are single, pointing to potential liquidity constraints that can reinforce a vicious cycle between unemployment and homelessness.

To better understand the mechanism connecting job loss to homelessness, I track the monthly income trajectory, combining earnings and unemployment-insurance benefits, of workers who *eventually* become homeless following an economic layoff, alongside their cumulative entry rate into homelessness. [Figure 10](#) displays the result.

Three features stand out. First, income remains constant in the months immediately following the layoff, reflecting severance payments plus the onset of unemployment insurance (UI) benefits. Second, income then declines steadily, with sharper drops around months 12, 18, and 24 corresponding to standard French UI potential-benefit-duration thresholds (which depend on prior tenure and age). Third, the cumulative entry into homelessness accelerates exactly as income falls. Together, these patterns are most consistent with a liquidity-driven mechanism: workers can sustain housing payments while income is decent, but as UI benefits exhaust and savings deplete, the rent burden becomes unsustainable.

To formally test whether the aggregate effect masks substantial heterogeneity, I interact the $Treat_i \times Post_t$ indicator with different socio-demographic and job characteristics. [Figure 11](#) (socio-demographic characteristics) and [Figure 12](#) (job characteristics) plot the interaction coefficients from this specification.

Three patterns emerge clearly. First, the contract length of the job matters. Workers with less than one year of tenure face dramatically elevated homelessness risk after displacement; workers with more than two years of tenure are partially protected. This is consistent with both a liquidity story (longer tenure means higher savings) and a UI story (longer tenure mechanically means longer potential benefit duration). Second, earnings level matters. Workers earning $\leq \text{€}2,000$ per month before displacement are at substantially elevated risk; workers earning $\geq \text{€}3,000$ are not. The protective effect of a daily high UI rate, defined as receiving more than $\text{€}60$, reinforces this interpretation. Third, household structure and education amplify risk. Single individuals, those without a high-school diploma, and those living in disadvantaged neighborhoods face higher risk. Married workers and older workers (≥ 53 years) are partially protected, while younger workers (< 25 years) face somewhat higher (though noisier) risk.

5.4 Robustness: Variation in the Timing of Job Displacement

To further evaluate the robustness of these findings, I employ the alternative identification strategy described in the research design above that leverages the staggered timing of economic layoffs and uses ‘not-yet-treated’ units as controls. [Figure B3](#) presents the estimated impact on the risk of homelessness using this approach. Despite the distinct methodological choices, the empirical patterns remain robust, further validating the causal link between job displacement and housing instability.

6 The Effect of Shelter Access on Labor Market Trajectories

To study the reverse direction of the vicious cycle, I turn to the labor market consequences of homelessness itself. The ideal experiment would randomly assign housing among otherwise identical workers who have just lost their home. I exploit the closest counterpart available in the shelter system: whether an individual’s very first request for shelter is met. With the shelter system being highly saturated, this depends on the leniency of the call-center agent who happens to pick up the call. Because calls are quasi-randomly assigned to agents within a location, month, and shift, agent leniency isolates a positive housing shock, that is rapid access to a shelter as opposed to at least some period of street homelessness. I find that rapid shelter access reduces job separations among callers who are employed and raises job finding among callers who are unemployed, which are precisely Assumptions 1 and 2 of the conceptual framework in [Section 4](#).

6.1 Empirical Framework

This section describes the instrumental variables approach used to estimate the causal effect of rapid shelter access on the labor market outcomes of individuals experiencing homelessness. I discuss the assumptions underlying the research design and provide evidence supporting them.

6.1.1 Sample Construction

I focus on individuals first request for shelter via the 115 hotline. I then make three further restrictions. First, because systematic call-level tracking only begins in 2020, I focus on everyone calling from this moment onwards. However, during the peak of the COVID-19 crisis, assignment rules differed from the regular regime as part of a large-scale effort to reduce street homelessness

through the requisitioning of hotels. I therefore start the sample in May 2021, the end of the extended post-COVID eviction moratorium, after which the additional hotel capacity was progressively withdrawn, in particular from 2022 onwards. Third, to observe labor market outcomes for at least one year after the call, I restrict the sample to calls made before March 2024. Finally, I restrict the sample to the working-age population, defined as individuals aged 18 to 60 at the time of the call.

The final sample consists of 96,755 individuals making their first request for shelter between May 2021 and March 2024. I consider two subpopulations separately: individuals who hold an employment contract at the time of the call (15,924 callers, the “employed” sample) and individuals without a contract at the time of the call but with a prior labor market history (81,461 callers, the “unemployed” sample).⁹

A natural concern is selection into who is calling the hotline. Several features of the setting alleviate this concern. First, the hotline is the main entry channel into the shelter system: it receives over 14 million calls per year ([Dihal, 2023](#)), and 95% of individuals experiencing homelessness in France pass through the shelter system. Second, because the system is saturated, the call records cover both individuals who are admitted to a shelter and those whose request is denied, so that the data also capture, at least in part, individuals who end up on the street. Nonetheless, interviews conducted during the annual street counts (*Nuit de la Solidarité*) indicate that many individuals sleeping rough have not called the hotline recently, consistent with discouragement after repeated denials ([pour le logement, 2025](#)). To address this form of selection directly, I restrict the analysis to the very first call an individual makes, before any discouragement can have occurred.

If anything, individuals who call are likely to be positively selected on dimensions such as mental health and substance use, since these conditions make it less likely that an individual is willing or able to call.¹⁰ Under random assignment of agents, this selection does not threaten internal validity; it does, however, matter for interpretation. The estimates below apply to the population of first-time callers, which is arguably the population most relevant for the labor market question at hand, and the complier population is characterized further in [Section 6.2](#).

⁹Restricting the unemployed sample to individuals with a prior labor market registry, in addition to the age restriction, brings the sample closer to the labor force. Together with the nationality controls introduced below, this also addresses the concern that some callers remain out of the labor force because they lack a work permit.

¹⁰In some cases, a third party (typically a social worker) places the request on the individual’s behalf; these cases represent X% of first calls and excluding them does not affect the results.

6.1.2 Instrumental Variables

The challenge for interpreting OLS in this setting is that whether a first request is met may be correlated with the caller’s unobservables or with the timing of unobserved shocks that affect subsequent labor market outcomes. The descriptive analysis in [Section 3](#) shows that many callers have experienced negative labor market shocks in the run-up to the call, which makes it especially difficult to disentangle the role of homelessness itself from the distress that precedes it. The aim here is therefore an instrument that is independent of these differences.

I leverage the quasi-random assignment of calls to call-center agents for identification. My instrument $Z_{j(i)}$ is the leave-one-out shelter-request acceptance rate of agent j who handles individual i ’s first call. I estimate the following population regression model using two-stage least squares:

$$S_i = \gamma Z_{j(i)} + X_i' \alpha + \psi_{c(i)} + v_i, \quad (11)$$

$$Y_i = \beta S_i + X_i' \delta + \psi_{c(i)} + \varepsilon_i, \quad (12)$$

where S_i is an indicator for whether individual i ’s first request for shelter is met, Y_i is the labor market outcome of interest, and X_i is a set of individual controls (age bins and nationality groups).¹¹ The fixed effects $\psi_{c(i)}$ are for the cell within which assignment to agents is as good as random: *département* $\times$ year-month $\times$ shift, and household type.¹² Household type distinguishes single men, single women, single men with children, single women with children, couples with children, couples without children, and other household configurations. Standard errors are clustered at the agent level.¹³ For this analysis I restrict the sample to calls handled by agents with at least 30 first calls in the data, so that the leniency measure can be constructed with reasonable precision.

If the IV assumptions hold and equations (11)–(12) are correctly specified ([Blandhol, Bonney,](#)

¹¹Nationality is recorded in the *SI-SIAO* for every caller, whether or not the record is linked to the employment data. This is why, unlike the layoff design in [Section 5](#), the analysis here does not need to be restricted to French nationals.

¹²The *SIAO* is organized at the *département* level (96 units, see [Section 2](#)), and so is the allocation of shelter places; this is therefore the level at which the instrument and the fixed effects are constructed. The shift fixed effects ensure that agents are compared to other agents working the same shift in the same *département* and month. This rules out that agents who systematically work later shifts, when fewer places are left, appear more stringent purely because of capacity. Note that shift fixed effects do not, by themselves, address non-random sorting of calls to agents *within* a shift; this is what the balance tests below examine.

¹³Following [Chyn, Frandsen and Leslie \(2025\)](#), [Table C7](#) in [Appendix C](#) shows that inference is robust to clustering at the agent-by-year and *département* levels.

Mogstad and Torgovitsky, 2026), the 2SLS estimand for β is a positively weighted average of the effect of rapid shelter access among compliers, that is, callers whose request would have been met had their call been picked up by a more lenient agent and denied had it been picked up by a less lenient one.

6.1.3 The Call-Center Agent Leniency Instrument

Although inspired by the judge stringency measures used in court settings to recover the causal effects of eviction (Collinson, Dutz, Humphries, Mader, Tannenbaum and van Dijk, 2026; Collinson et al., 2024), the leniency measure here differs in one important respect. If shelter places were freely available, agent leniency would not matter: every request would be met. The instrument therefore relies on the fact that agents face an overcrowded system in which they cannot admit everyone. Priority rules govern part of the allocation — families with small children and victims of domestic violence are prioritized — but within these rules agents handle requests differently. The margin of discretion is largest for single men, for whom no priority rule applies, and it is within this group that leniency varies most across agents in the same *département*, month, and shift.¹⁴

I construct agent leniency as the monthly leave-one-out mean acceptance rate of first requests made by single men and handled by the agent picking up the call, using all 115 call records rather than only the linked sample.¹⁵ Agent leniency is strongly predictive of whether a caller’s first request is met. Figure C1 shows the distribution of agent leniency, residualized by *département*-year-month-shift cell, across calls. The variation is substantial: a 11.7 percentage point difference in the probability that a request is met separates the 10th from the 90th percentile of agent leniency.

6.1.4 Validity of the IV Design

I next discuss tests of the assumptions under which agent leniency is a valid instrument and the IV estimand reflects a positively weighted average of local treatment effects on compliers.

Relevance. To assess the relevance of the instrument, Table C2 reports first-stage estimates

¹⁴To fix ideas, consider a single man reaching the hotline early in the day, when some places are still available. One agent may assign a place directly; another may hold it back for callers with higher priority and ask the man to call back at the end of the day in case places remain. Many do not call back, which generates variation in shelter versus street exposure that is unrelated to the caller’s characteristics.

¹⁵Over the period 2018–2026, 6,954 distinct agents appear in the call records; 3,583 handle calls between May 2021 and March 2024, of whom 2,457 handle at least 30 first calls from single men.

from equation (11) for the employed and the unemployed samples, first with cell fixed effects only and then adding the individual controls. Agent leniency has a large and statistically significant effect on the probability that the first request is met. The Kleibergen–Paap F -statistics are 666.6 in the employed sample and 1,413.8 in the unemployed sample, providing evidence against weak instruments. The first stage is essentially unchanged when adding controls, a first indication that leniency is uncorrelated with individual characteristics.

Exogeneity. Figure C2 presents evidence that socio-demographic characteristics and the caller’s current and past labor market situation are not predictive of agent leniency, which lends empirical support to the quasi-random assignment of calls to agents. Panel (a) reports, for the employed sample, the coefficients from regressions of leniency on age, gender, nationality, and the characteristics of the current contract (earnings, contract length, contract type, full-time status, and occupation group), all expressed in standard deviation units; panel (b) repeats the exercise for the unemployed sample using past labor market history (days worked in the previous one and two years, time since the last job, and the earnings and contract type of the last job). The coefficients are uniformly small, with the exception of modest imbalances in nationality and age, which I include as controls in all specifications. An F -test that all coefficients are jointly equal to zero cannot reject the null in either sample ($p = X$ and Y).

Monotonicity. The monotonicity assumption of Imbens and Angrist (1994) requires that callers whose request is met would also have had it met by a more lenient agent, and that callers whose request is denied would also have had it denied by a less lenient agent. Frandsen, Lefgren and Leslie (2023) show that a weaker, average monotonicity condition is sufficient for IV to have a causal interpretation in randomized examiner designs. It would violate average monotonicity, for example, if agents who are lenient with single men were stricter than average with other callers. This concern is more acute here than in court settings, because shelter places are rival within a shift: a place given to one caller is not available to the next. Two features of the setting nonetheless support the assumption. First, because the hotline is heavily used, agents collect only a minimum of information to assess priority — household composition and the stated reason for the call — so that the dimension along which they can rank callers differently is narrow. Second, I test the assumption in two ways (Table C3). I first estimate the first stage separately by household type, gender, age group, nationality group, and employment status at the time of the call; the first-stage estimates are positive in every subsample and similar in magnitude to the main estimate. I then follow Bhuller, Dahl, Løken and Mogstad (2020) and Norris, Pecenco and Weaver (2021)

and construct leniency from one subpopulation — single men, the group for which no priority rule applies — and use it as the instrument for each other household type. An agent who is lenient with single men is also lenient with single women, single parents, and couples: the first stage is positive and of similar size in every group, and constructing leniency within household type instead yields first stages that are very similar to the baseline. Agents thus move all callers in the same direction, which is consistent with average monotonicity holding in this setting fill in the first-stage coefficients by subgroup in the table.

Exclusion. A key assumption that cannot be tested directly is that agent leniency affects callers’ labor market outcomes only through whether the first request for shelter is met. The setup of the hotline limits the scope for violations: the agent’s role is to allocate a place according to priority rules, and the agent has no further contact with the caller once the place is assigned or denied. Due to the high saturation of the call center, agents spend only minimal time on the phone with unhoused individuals, making it very unlikely that they impact their trajectory other than through their decision to grant the shelter bed.

6.2 Estimates of the Causal Effect of Shelter Access

In this section I present the main estimates of the effect of rapid shelter access on labor market outcomes. The variation in whether the first request is met allows me to isolate a positive housing shock at the moment of entry into homelessness: the comparison is between individuals who are directly sheltered — a precarious but stable housing situation — and individuals who spend at least some period without shelter because their request is denied.¹⁶ This setup allows me to test the two assumptions of the conceptual framework: that homelessness raises job separation (Assumption 1, $s_N > s_H$) and lowers job finding (Assumption 2, $f_N < f_H$). I first present the effects on job separation, restricting the analysis to callers employed at the time of the call, and then the effects on job finding for callers unemployed at the time of the call. I then turn to mechanisms.

¹⁶This interpretation assumes that the caller has no outside option other than the street. Based on the housing situation reported at the time of the call, this holds for at least 50% of callers, who report sleeping rough, and for a further 15% who report being in an emergency shelter that they must leave VERIFY: exact shares of reported housing situation at the first call, from the emergency and insertion request data.

6.2.1 Job Separation

Table 3 reports the effects of rapid shelter access on job separation for the employed sample. Column (1) shows that workers whose first request is met are 7.2 percentage points less likely to see their contract end within three months of the call, a 17% reduction relative to the control mean of 42.1%. Column (2) shows the mirror image over a longer horizon: they are 6.5 percentage points, or 20%, more likely to still be employed in the same contract one year after the call.

Columns (3) to (7) decompose the contract endings by the reason recorded by the employer in the *DSN*. Rapid shelter access reduces contract terminations — as opposed to the scheduled end of a fixed-term contract — by 8.4 percentage points, or 33% of the control mean. Two channels account for this reduction: sheltered workers are less likely to quit their position (a 31% reduction relative to the control mean of 12.5%) and less likely to be dismissed at the end of their trial period (a 64% reduction relative to a control mean of 3.9%). Dismissals for misconduct also decline, though the estimate is imprecise, and there is no effect on contract endings for health reasons. The pattern is consistent with stable shelter allowing workers to keep meeting the day-to-day demands of a job — showing up, and showing up on time — during the most disruptive weeks of a housing loss, and with quits reflecting the difficulty of holding a job while unsheltered rather than a move to a better job.

6.2.2 Job Finding

Table 4 reports the effects on job finding for the unemployed sample. Rapid shelter access raises the probability of being employed one year after the call by 6.0 percentage points, a 74% increase over a control mean of 8.1%. The low control mean is itself informative: it documents that most unemployed individuals entering homelessness remain out of formal employment a year later. The effect is not confined to marginal jobs. The probability of holding an open-ended contract one year after the call rises by 1.6 percentage points (46% of the control mean), the number of days worked during the first year rises by 16 days (65% of the control mean), and unconditional monthly earnings one year after the call rise by €72, or 76%.

Taken together, the separation and finding results establish the reverse direction of the vicious cycle: the loss of shelter both destroys existing employment relationships and impedes the formation of new ones.

6.3 Mechanisms

The large effects of a single met request may appear surprising at first: a shelter provides a roof, but individuals remain in a precarious housing situation. This section provides evidence on what lies behind the estimates: subsequent engagement with homelessness services and entry into transitional housing units. It also examines heterogeneity across callers.

6.3.1 Engagement with Homelessness Services

[Table 5](#) pools the employed and unemployed samples and examines how a met first request shapes subsequent engagement with the shelter system. Individuals whose first request is met spend 50 more nights in shelters during the year following the call, relative to a control mean of only 7 nights (column 1). The decomposition in columns (3) and (4) shows that most of these additional nights are spent in emergency shelters (45 nights), but 6 additional nights are spent in transitional housing, and the probability of entering transitional housing within a year doubles, from 3.6% to 7.2% (column 2).

These results confirm the discouragement mechanism underlying the instrument. Individuals whose first request is denied can call again, yet they spend far fewer nights in shelters over the following year, so that a single denied request translates into substantially longer exposure to street homelessness. They also indicate that the treatment identified by the leniency instrument is a bundle: rapid access to emergency shelter, more nights sheltered overall, and faster entry into transitional housing, which in turn comes with social work support.

6.3.2 The Role of Transitional Housing Shelters

To disentangle the role of rapid access to longer-term transitional housing from that of emergency shelter, I use a second instrument that exploits congestion in the transitional-housing system at the time of the request, measured by the leave-one-out mean waiting time until a first offer within the *département* $\times$ household type $\times$ application-month cell. TO ADD: waiting-time IV results; the balance check in [Figure C3](#) shows that past labor market history does not predict waiting times, with the exception of a modest imbalance in fixed-term contract status of the last job. TO ADD: effects for transitional housing shelters

6.3.3 Heterogeneity

TO ADD: heterogeneity by gender, age, nationality group, prior labor market attachment, contract type at request (open-ended vs. fixed-term), and *département*-level saturation. Given the framework, the separation effect should be concentrated among workers with short tenure (trial period) and the finding effect among those with recent attachment.

6.4 Robustness

TO ADD: all robustness checks

Finally, Appendix E presents an alternative design that exploits a negative housing shock — landlords reclaiming their property for personal use or sale — and the staggered timing of entry into homelessness following this event. This design identifies the effect of housing loss on a selected sample of individuals who eventually enter the shelter system, but it does not rely on the shelter allocation process at all. It yields the same qualitative conclusion: sharp and persistent declines in employment and earnings following the loss of housing.

7 Quantifying the Vicious Cycle

The two quasi-experiments identify the two directions of the cycle separately. This section combines them in the four flow equations of Section 4 to answer three questions: how much does homelessness contribute to unemployment among low-income workers; how much of that cost is insured, given that the shelter system is rationed; and what would it cost to insure the rest.

8 Conclusion

still to adjust

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Tables

Table 1: Sample Characteristics of Matched vs Non-Matched Homeless Individuals

	Not Matched		Matched		Δ		
	$\bar{x}$	sd	$\bar{x}$	sd	$\Delta\bar{x}$	Δ sd	p-val
Panel A: Information about Request							
Year of Request	2021	2.506	2020	2.615	-0.519	0.003	0.000
Household Type:							
Couple with Children	0.085	0.279	0.066	0.248	-0.019	0.000	0.000
Couple without Children	0.027	0.163	0.025	0.155	-0.003	0.000	0.000
Single Women	0.139	0.346	0.165	0.371	0.026	0.000	0.000
Single Women with Kids	0.120	0.325	0.140	0.347	0.020	0.000	0.000
Group with Kids	0.135	0.342	0.101	0.302	-0.034	0.000	0.000
Adults without Kids	0.071	0.257	0.062	0.241	-0.009	0.000	0.000
Isolated Man	0.390	0.488	0.421	0.494	0.031	0.001	0.000
Man with Kids	0.012	0.107	0.015	0.121	0.003	0.000	0.000
Panel B: Socio-demographic Characteristics							
Age	35.839	19.838	36.515	12.470	0.676	0.017	0.000
Female	0.413	0.492	0.425	0.494	0.012	0.001	0.000
French Nationality	0.298	0.458	0.535	0.499	0.236	0.001	0.000
Observations	1,118,749		1,044,769		2,163,518		

Notes: This table compares the baseline characteristics of individuals who submitted a shelter request and were successfully linked to administrative labor market records against those who remained unmatched. The sample is restricted to adults without a recent migratory history; this exclusion is necessitated by the fact that both minors and recent arrivals to France have a low probability of appearing in national employment and unemployment registries, which would lead to mechanical attrition in the matching process.

Table 2: The Effect of Job Loss on Labor Market Outcomes and Homelessness Risk

	Employment	Employment (Open-Ended)	Earnings	Probability of Homelessness
	(1)	(2)	(3)	(4)
Economic Layoff	-0.32771*** (0.002)	-0.41779*** (0.002)	-661.46718*** (8.890)	0.00025** (0.000)
Mean outcome, treated at t= 0	1	1	2,109	0.0005
Effect relative to mean	-32%	-41%	-31%	50%
Number of obs.	2,681,175	2,681,175	2,681,175	2,681,175

Notes: This table reports the estimated impact of an economic layoff event on labor market outcomes and the risk of homelessness, utilizing the matched difference-in-differences framework specified in Equation 10. The treatment group comprises workers displaced in economic layoffs of two or more workers in a given firm-month between 2017 and 2023. The control group consists of matched workers who were not employed at firms engaged in economic layoffs during the analysis period and remained employed during the displacement year. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Rapid Shelter Access Reduces Job Separation

	Employed at Request						
	Contract End ≤ 3 Months	Contract End ≥ 1 Year	Contract Termination	Reasons Contract End			
				Misconduct	Quit	End Trial Period	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
First Request Met	-0.072** (0.034)	0.065* (0.036)	-0.084*** (0.030)	-0.027 (0.016)	-0.039* (0.023)	-0.025* (0.013)	-0.000 (0.008)
Control mean	0.421	0.333	0.257	0.081	0.125	0.039	0.015
Effect relative to mean	-17%	20%	-33%	-33%	-31%	-64%	0%
Kleibergen–Paap F	666.59	666.59	666.59	666.59	666.59	666.59	666.59
Observations	15,924	15,924	15,924	15,924	15,924	15,924	15,924

Notes: This table reports 2SLS estimates of equation (12) for individuals employed at the time of their first request for shelter. The endogenous variable is an indicator for the first request being met, instrumented by the leave-one-out acceptance rate of the call-center agent handling the call. All specifications include *département* $\times$ year-month $\times$ shift and household-type fixed effects, and controls for age bins and nationality groups. Column (1): contract ended within 3 months of the call. Column (2): still employed in the same contract 12 months after the call. Column (3): contract terminated define: termination initiated by employer or employee, as opposed to scheduled expiry of a fixed-term contract; horizon. Columns (4)–(7) decompose (3) by the reason recorded in the DSN. Standard errors clustered at the agent level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Rapid Shelter Access Raises Job Finding

	Unemployed at Request			
	Job (1 Year)	Open-ended contract (1 Year)	Days Worked (1 Year)	Earnings (1 Year)
	(1)	(2)	(3)	(4)
First Request Met	0.060*** (0.012)	0.016** (0.007)	15.664*** (3.482)	71.832*** (14.532)
Control mean	0.081	0.035	24.035	93.990
Effect relative to mean	74%	46%	65%	76%
Kleibergen–Paap F	1,413.83	1,413.83	1,413.83	1,413.83
Observations	81,461	81,461	81,461	81,461

Notes: This table reports 2SLS estimates of equation (12) for individuals without an employment contract at the time of their first request for shelter but with a prior registration in the employment or unemployment records. The endogenous variable is an indicator for the first request being met, instrumented by the leave-one-out acceptance rate of the call-center agent handling the call. All specifications include *département* $\times$ year-month $\times$ shift and household-type fixed effects, and controls for age bins and nationality groups. Column (1): employed 12 months after the call. Column (2): holds an open-ended contract 12 months after the call. Column (3): days worked during the 12 months following the call. Column (4): unconditional monthly earnings (euros) 12 months after the call. Standard errors clustered at the agent level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

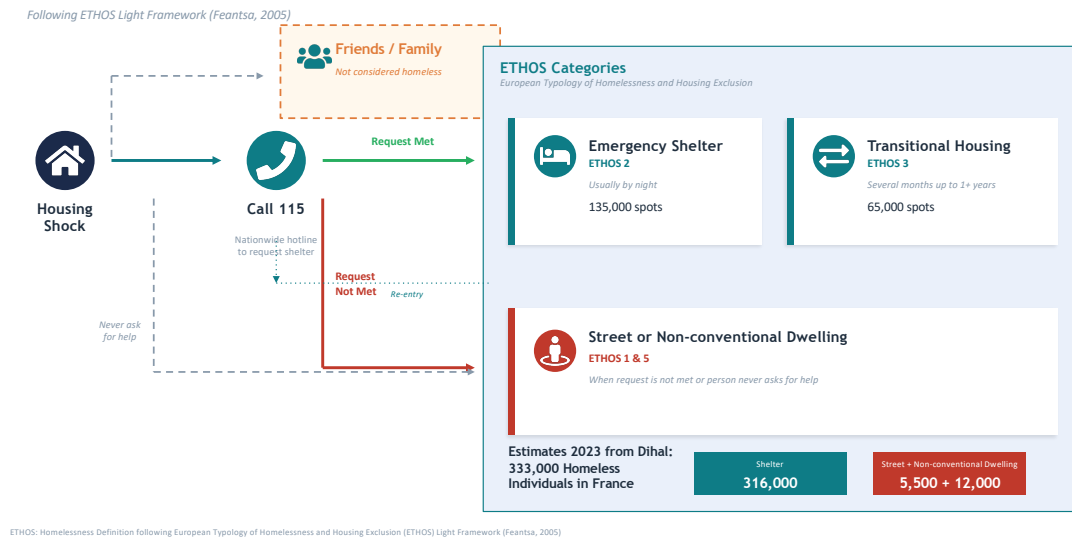
Table 5: Mechanism: Rapid Shelter Access and Subsequent Use of Homelessness Services

	Labor Force at Request			
	Nights in Shelter	Transitional housing	Nights	Nights
	(12 months)	(12 months)	Emergency Shelter	Transitional Housing
	(1)	(2)	(3)	(4)
First Request Met	50.404*** (5.278)	0.036*** (0.011)	44.583*** (5.440)	6.225*** (2.074)
Control mean	6.703	0.036	2.831	3.904
Kleibergen–Paap F	1,341.62	1,341.62	1,341.62	1,341.62
Observations	97,385	97,385	97,385	97,385

Notes: This table reports 2SLS estimates of equation (12) for the pooled sample of employed and unemployed callers. Column (1): total nights spent in any shelter during the 12 months following the call. Column (2): indicator for entering transitional housing within 12 months. Columns (3) and (4) decompose column (1) into nights in emergency shelter and nights in transitional housing. Specification and standard errors as in Table 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

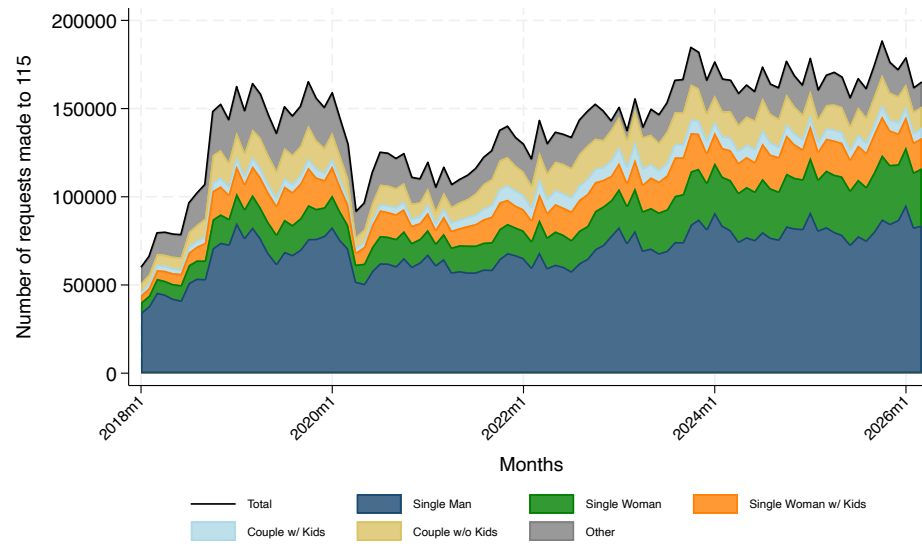
Figures

Figure 1: Pathways After a Housing Shock in France



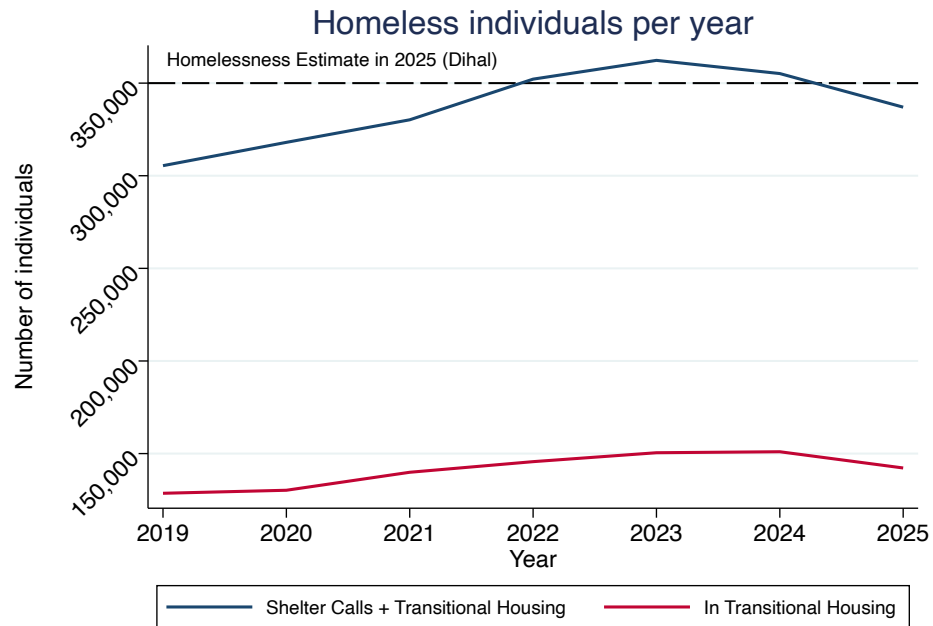
Notes: Following the ETHOS Light Framework (Feantsa, 2005). Individuals experiencing a housing shock may call the 115 hotline; requests are met (emergency shelter or transitional housing) or unmet (street/non-conventional dwelling). Some never call (stay with family/friends, or remain on the street). Estimates from Dihal (2022): of 333,000 homeless individuals in 2023, 316,000 are in shelters and 5,500–12,000 on the street or in non-conventional dwellings.

Figure 2: Calls to 115 by Household Situation Over Time



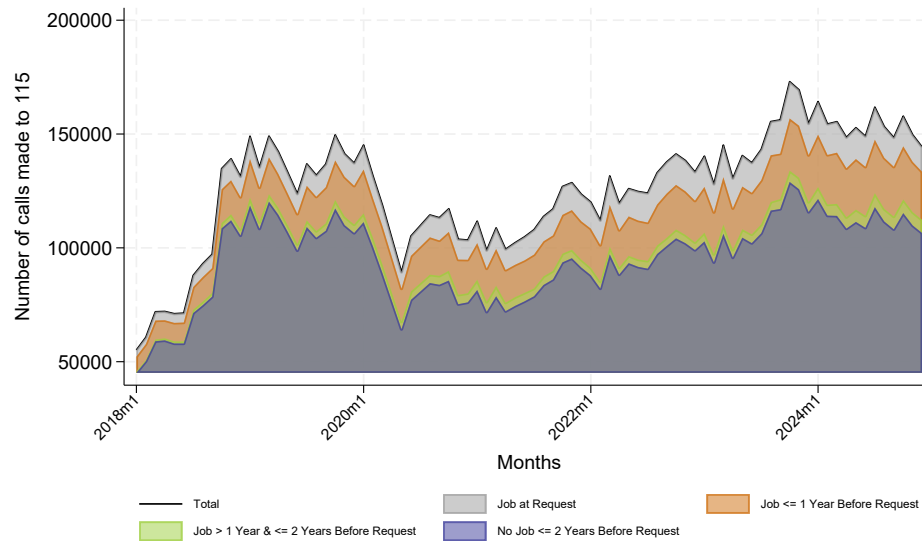
Notes: Monthly count of requests for shelter to the 115 hotline, decomposed by household type of the caller. Source: SI-SIAO administrative records, 2018–2026.

Figure 3: Coverage of the SI-SIAO Data vs. Dihal Estimate



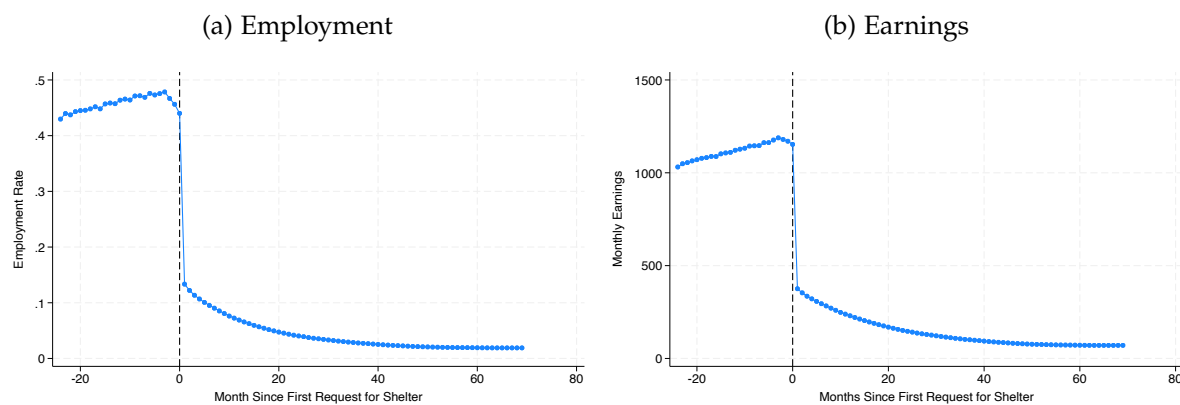
Notes: Annual count of unique individuals appearing in SI-SIAO (shelter calls and transitional housing) compared to the Dihal estimate of the total homeless population in France. The series converge by 2022, indicating that SI-SIAO records capture essentially the entire sheltered population.

Figure 4: 115 Calls Decomposed by Labor-Market Attachment



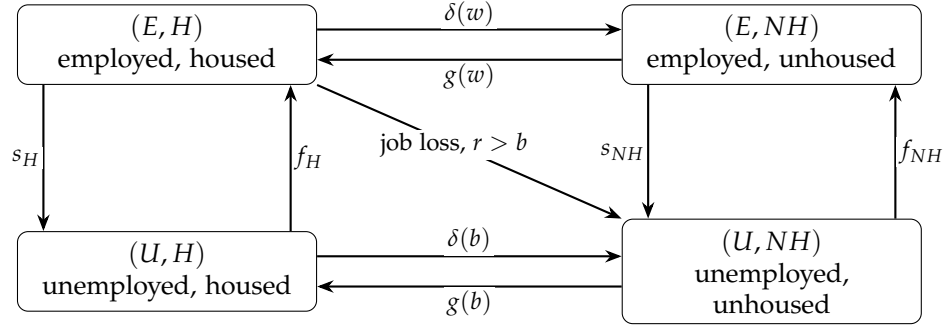
Notes: Monthly count of requests for shelter to the 115 hotline, decomposed by the caller's labor-market status at the time of the call. "Job at request" means an active employment contract on the day of the call. "Job $\leq$ 1 year before" means at least one work contract within the prior 12 months but none active on the day of the call. Source: Own calculations, based on SI-SIAO merged with DSN administrative records, 2018–2024.

Figure 5: Earnings and Employment Trajectories Around Entry Into Homelessness



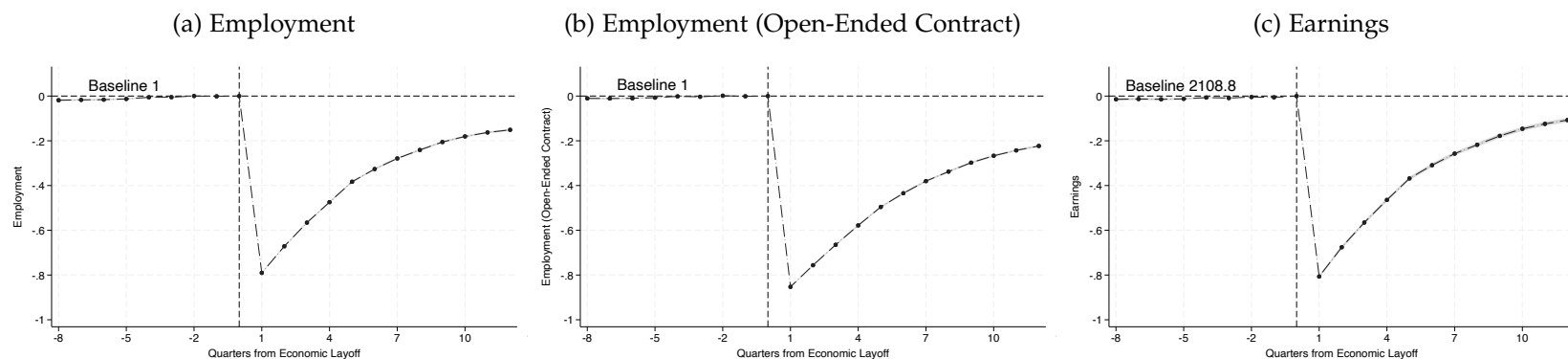
Notes: This figure illustrates the evolution of employment and earnings trajectories for individuals following their initial transition into homelessness. The onset of homelessness is proxied by the date of the individual's first request for shelter. The sample is restricted to individuals who report at least a marginal labor market attachment, defined as having been employed for a minimum of 180 days within the five years preceding the initial shelter request.

Figure 6: Monthly transitions between employment and housing states



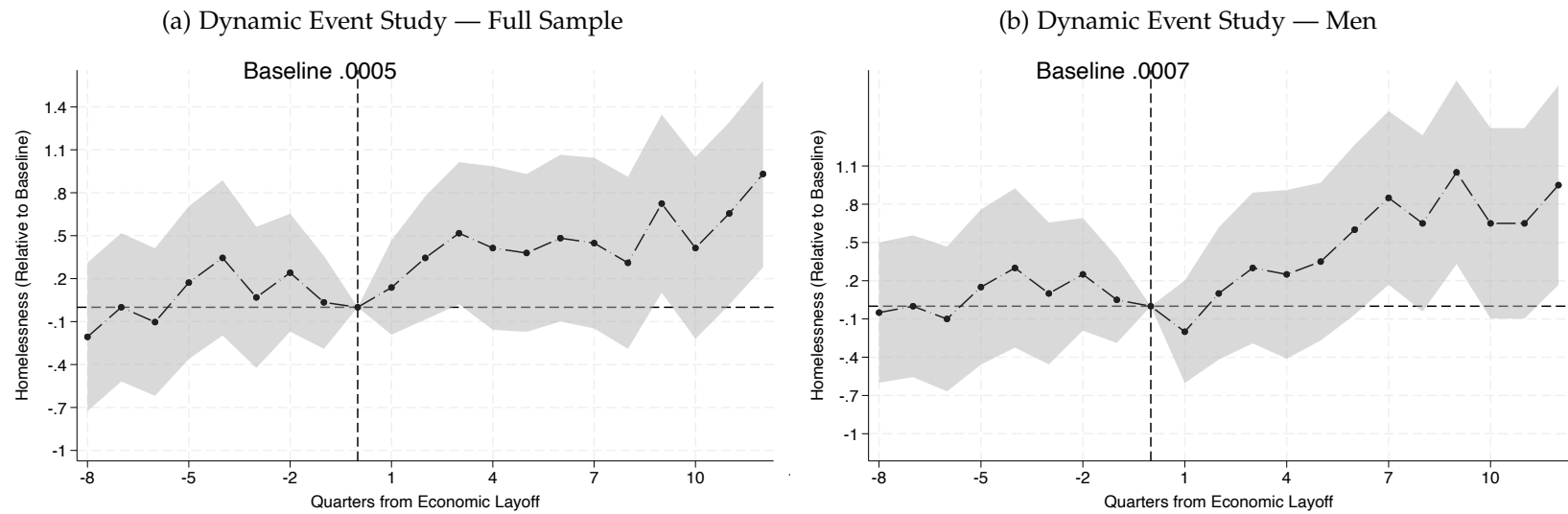
Notes: Arrows show the monthly transitions of the model and their probabilities. s_H , f_H and s_{NH} , f_{NH} are the separation and job-finding probabilities of housed and unhoused workers; $\delta(y)$ and $g(y)$ are the probabilities of losing and regaining housing at income y . The diagonal arrow is the loss of housing at job loss for workers whose rent exceeds benefits. Several transitions can occur in the same month.

Figure 7: The Effect of Job Loss on Labor Market Outcomes — Relative to Baseline



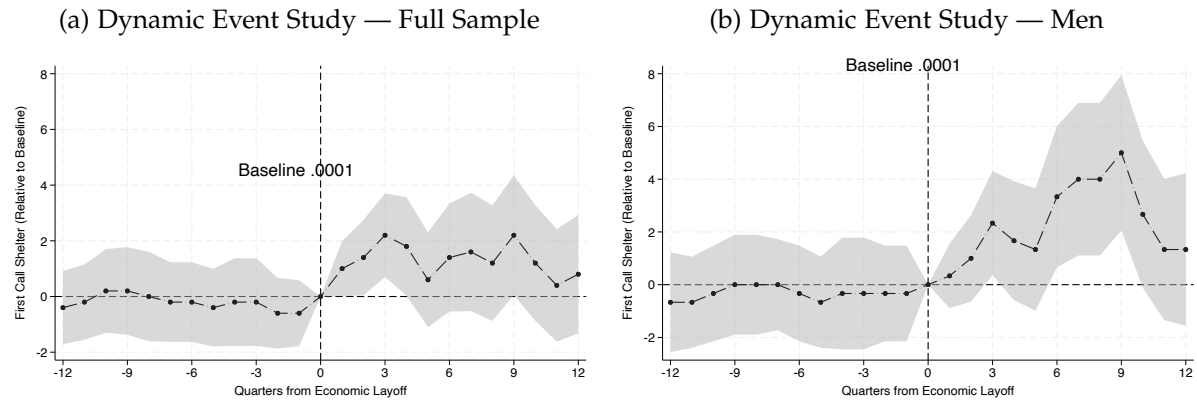
Notes: This figure presents the estimated impact of job loss on labor-market outcomes using the matched difference-in-differences specification in Equation 9. Estimates are normalized relative to the baseline mean of the outcome at the time of displacement. The horizontal axis represents quarters relative to the economic layoff event. The treatment group comprises workers displaced in an economic layoff (at least two workers in a given firm-month for economic reasons) between 2017 and 2023. The control group consists of matched workers who were not employed at firms engaged in economic layoffs during the analysis period and remained employed during the displacement year. Shaded areas denote 90% confidence intervals.

Figure 8: The Effect of Job Loss on Homelessness Risk — Relative to Baseline



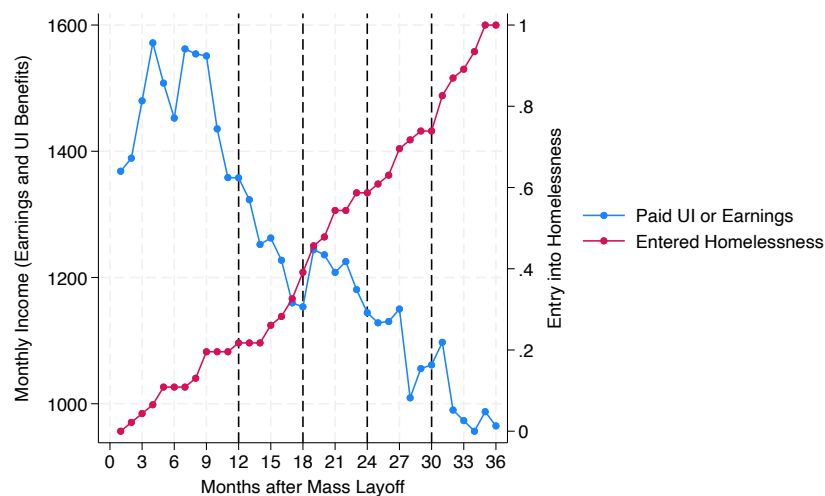
Notes: This figure presents the dynamic impact of an economic layoff on the probability of entering homelessness using the matched difference-in-differences specification in Equation 9. Estimates are normalized relative to the baseline mean of the outcome at the time of displacement. The horizontal axis represents quarters relative to the economic layoff event. Left panel: full matched sample. Right panel: male workers only. Shaded areas denote 90% confidence intervals. Homelessness rates approximately double relative to baseline three years after the economic layoff event.

Figure 9: Timing of Entry into Homelessness: Effect of Job Loss on the First Call for Shelter



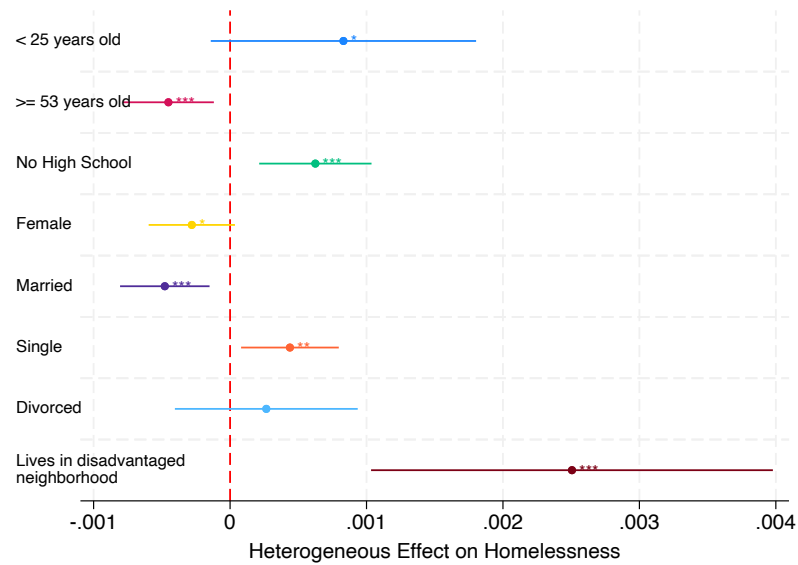
Notes: This figure presents the dynamic impact of an economic layoff on the probability of making a *first* call for shelter, using the matched difference-in-differences specification in Equation 9. Estimates are normalized relative to the baseline mean (0.01%). The horizontal axis represents quarters relative to the economic layoff event. Shaded areas denote 90% confidence intervals. The increase in first calls is concentrated one to two years after the layoff, consistent with the exhaustion of unemployment insurance benefits.

Figure 10: Income and Cumulative Entry into Homelessness After an Economic Layoff



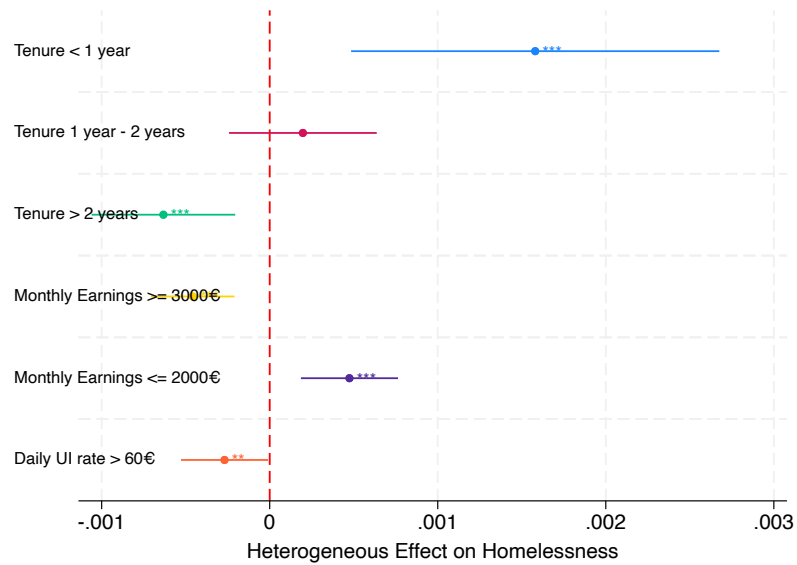
Notes: Sample restricted to workers who experience an economic layoff and subsequently enter homelessness. The left axis (blue) plots average monthly income, combining earnings and unemployment insurance benefits. The right axis (red) plots the cumulative share who have entered homelessness. The horizontal axis is months after the layoff event. Income peaks 4–6 months after displacement (severance plus UI onset), declines as UI benefits exhaust, and homelessness entry accelerates exactly as income falls.

Figure 11: Heterogeneity Analysis: Sociodemographic Characteristics



Notes: This figure plots the interaction coefficients from a specification that augments Equation 10 with interactions between the treatmentxpost indicator and worker characteristics. Outcome variable: probability of homelessness. Confidence intervals are 95% and based on standard errors clustered at the individual level. Household composition, age, and education are all significantly related to the homelessness response to job loss.

Figure 12: Heterogeneity Analysis: Job Characteristics

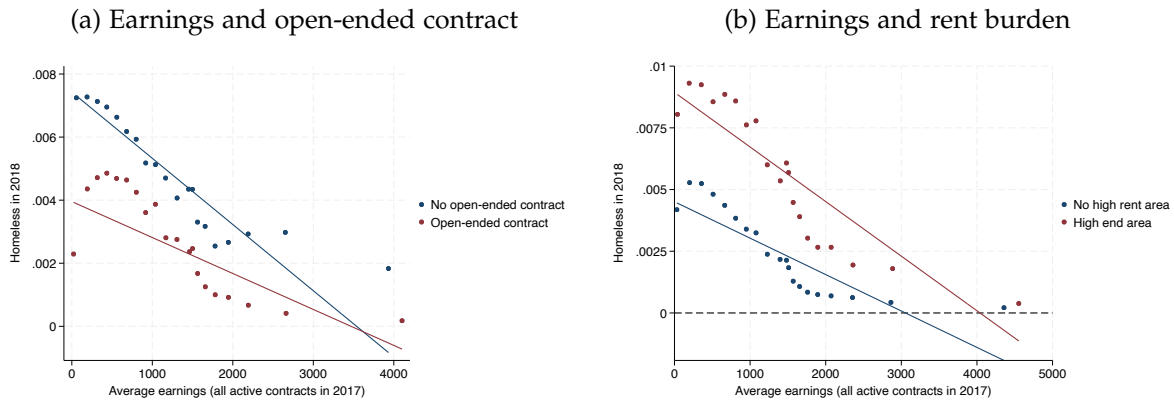


Notes: This figure plots the interaction coefficients from a specification that augments Equation 10 with interactions between the treatmentxpost indicator and worker characteristics. Outcome variable: probability of homelessness. Confidence intervals are 95% and based on standard errors clustered at the individual level. Short tenure and low earnings sharply elevate homelessness risk after job loss; a high UI replacement rate provides partial protection.

A Data and Sample Construction

Still add more details for sample construction here

Figure A1: Homelessness Risk and Precarious Labor Market Conditions



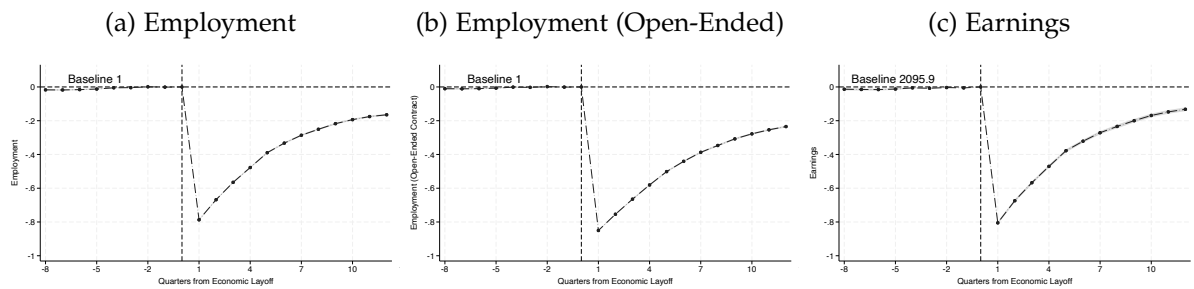
Notes: Binned scatter plots of the probability of a homelessness event in 2018 against average monthly earnings in 2017, for individuals registered with France Travail between 2017 and 2025 and holding an active employment contract in 2017. Panel (a) splits the sample by whether the individual holds an open-ended contract in 2017; panel (b) by whether the individual lives in a high-rent area define high-rent area. Source: EALE slide 9.

B The Effect of Job Loss on Homelessness: Additional Results

This appendix reports robustness checks and additional heterogeneity results for the matched difference-in-differences design of [Section 5](#).

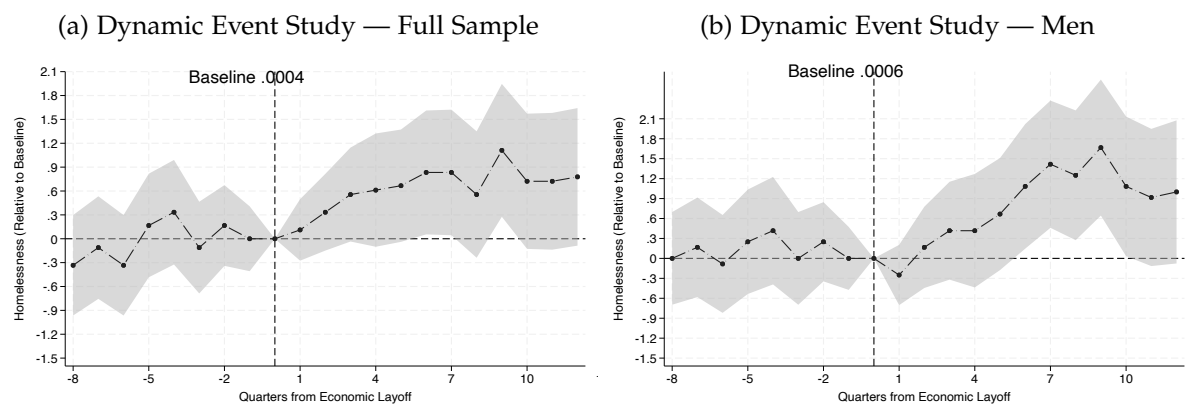
B.1 Robustness

Figure B1: Robustness — Excluding 2020 and 2021: Effect on Labor Market Outcomes (Relative to Baseline)



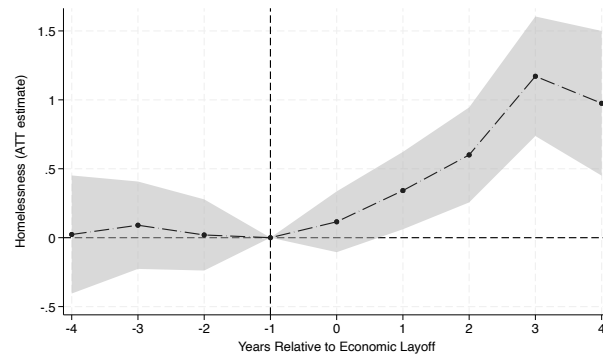
Notes: Robustness check excluding economic layoffs occurring in 2020 and 2021 to ensure that the main findings are not driven by the COVID-19 pandemic and associated labor-market disruptions. All other specification choices follow the main analysis.

Figure B2: Robustness — Excluding 2020 and 2021: Effect on Homelessness (Relative to Baseline)



Notes: Robustness check excluding economic layoffs occurring in 2020 and 2021. All other specification choices follow the main analysis. Left panel: full matched sample. Right panel: men only.

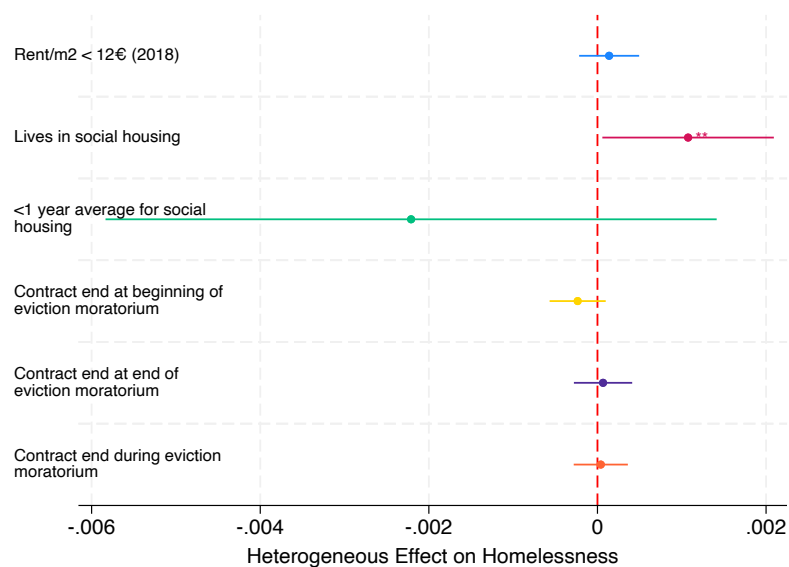
Figure B3: Robustness — Only Treated: Effect on Homelessness (Relative Effects)



Notes: Robustness check restricting the sample to treated workers only and using staggered-timing variation to identify dynamic effects on homelessness. All other specification choices follow the main analysis.

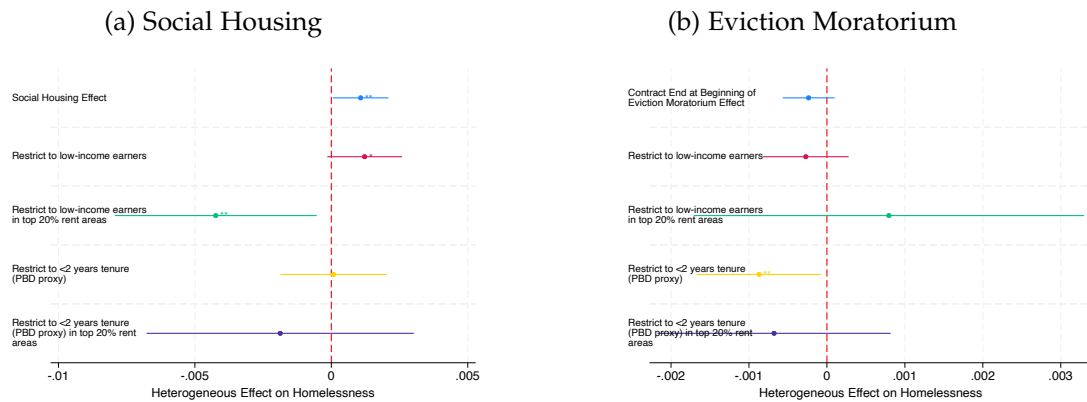
B.2 Heterogeneity: Housing Policies

Figure B4: Heterogeneity Analysis: Housing Policies



Notes: This figure plots the interaction coefficients from a specification that augments Equation 10 with interactions between the treatment indicator and pre-displacement housing-policy characteristics. Outcome variable: probability of homelessness. Variables include indicators for rent per square meter below €12 in 2018, residence in social housing, short average wait time for social housing, and whether the worker's contract ended at the beginning, end, or during an eviction moratorium period. 95% confidence intervals based on standard errors clustered at the plant level.

Figure B5: Heterogeneity Analysis: Combination of Housing and Income



Notes: This figure plots heterogeneous treatment effects on homelessness combining housing policy variables with income/tenure restrictions. Left panel: social housing interacted with low-income, high-rent area, and short-tenure (PBD proxy) subsamples. Right panel: contract-end timing relative to the eviction moratorium interacted with the same subsamples. 95% confidence intervals based on standard errors clustered at the plant level.

C The Effect of Shelter Access: Validity Checks and Additional Results

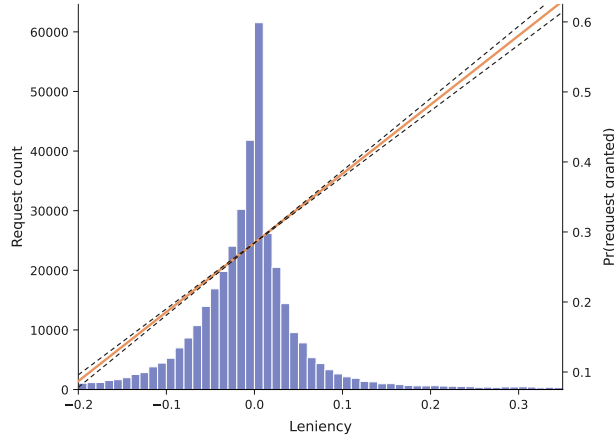
This appendix reports the validity checks and robustness results for the leniency design of [Section 6](#).

C.1 Sample and Instrument

Table C1: Summary Statistics: Employed and Unemployed Samples at First Request

TABLE TO ADD. Rows: age, female, nationality group, household type, reported housing situation at call, request met; labor market: contract type, earnings, days worked in past 1 and 2 years, days since last job. Columns: employed sample, unemployed sample, all linked first-time callers, all first-time callers (unlinked).

Figure C1: Distribution and First Stage of Call-Center Agent Leniency



The figure plots the distribution of the leave-out agent leniency instrument together with the first stage it generates. Blue bars show the number of requests in each of 55 equally spaced bins of width 0.010, measured against the left vertical axis; each request is placed at the leniency of the agent who handled it, so an agent with a large caseload contributes many requests at a single point. The solid orange line, plotted against the right vertical axis, gives fitted values from the first-stage regression of placement on leniency with cell fixed effects, and dashed lines give the corresponding 95% confidence interval. Black markers are means of the residualised placement rate within twenty-two equal-sized bins of the instrument, re-centred at the sample placement rate; they are shown to document that the first-stage relation is close to linear across the whole support of the instrument rather than only near its centre. The plotted range covers 98.3% of requests in panel (a) and 98.4% in panel (b); the remainder lie in the right tail and are almost entirely agents with small caseloads, for whom the leave-out mean is estimated imprecisely.

C.2 Validity of the IV Design

Table C2: First Stage: Agent Leniency and the Probability that the First Request is Met
TABLE TO ADD. Columns: (1) employed sample, cell FE only; (2) employed sample, + controls;
(3) unemployed sample, cell FE only; (4) unemployed sample, + controls. Rows: coefficient on
leniency, SE (agent cluster), control mean of request met, Kleibergen–Paap F (666.59 employed;
1,413.83 unemployed; 1,341.62 pooled), N (15,924; 81,461; 97,385).

Figure C2: Balance Check: Caller Characteristics and Agent Leniency

(a) Employed at request

(b) Unemployed at request



Notes: Each point is the coefficient from a regression of residualized agent leniency on the characteristic listed, expressed in standard deviation units, with 95% confidence intervals based on standard errors clustered at the agent level. Panel (a): demographics and characteristics of the current contract at the time of the first request (EALE appendix slide 21). Panel (b): demographics and past labor market history. Joint F -test p -values: X (a), Y (b).

Figure C3: Balance Check: Caller Characteristics and Leave-One-Out Waiting Times for Transitional Housing



Notes: Coefficients from regressions of the leave-one-out mean waiting time in the *département* $\times$ household type $\times$ application-month cell on the characteristic listed, in standard deviation units, with 95% confidence intervals. EALE appendix slide 22.

Table C3: Monotonicity Checks: First Stage by Subsample and with Reverse-Sample Instruments
TABLE TO ADD (results already estimated). Panel A: first stage by household type, gender, age group, nationality group, and employment status at request. Panel B: reverse-sample test — leniency constructed from single men used as instrument for each other household type; and leniency constructed within household type. Report coefficient, SE (agent cluster), KP F , N per row.

Table C4: Exclusion: Alternative Dimensions of Agent Discretion
TABLE TO ADD, if the SI-SIAO records allow it: correlation between baseline leniency and agent-level rates of (i) orientation to a social worker / day center, (ii) opening of a transitional-housing application, (iii) call-back advice; and 2SLS estimates controlling for these measures.

C.3 Robustness

Table C5: Robustness: Alternative Constructions of the Instrument

TABLE TO ADD. Columns: (1) baseline: single-men leave-one-out 2SLS with cell FE and controls; (2) same, cell FE only; (3) within-household-type leave-one-out; (4) UJIVE with agent indicators as instruments and the same cell FE (ManyIV package). Rows: contract end ≤ 3 months (employed sample), job at 1 year (unemployed sample), nights in shelter (pooled).

Table C6: Robustness: Excluding Non-Saturated *Départements*

TABLE TO ADD. Drop *département* $\times$ months in which the share of requests met exceeds X%.

Table C7: Robustness: Alternative Levels of Clustering

TABLE TO ADD. Standard errors clustered at agent, agent $\times$ year, and *département* level.

D Conceptual Framework

E Alternative Design: Housing Loss Following Landlord Repossession

This appendix presents an alternative design for the effect of housing loss on labor market trajectories that does not rely on the shelter allocation process. It focuses on workers who become homeless after their landlord reclaims the property for personal use or sale (*congé pour reprise ou vente*), and exploits variation in the timing of this housing shock. Using workers who experience the same shock only in later years as not-yet-treated controls, I document sharp and persistent declines in employment and earnings for those active in the labor market at the time of the shock.

E.1 Research Design

The ideal design would mirror the layoff design of [Section 5](#): a housing shock independent of the worker’s own income and rent history, observed for everyone who suffers it. Landlords reclaiming their flat for sale or personal use is such a shock, since, unlike an eviction for arrears, it is exogenous to the tenant’s rent payment history and thus to their labor market history. The limitation is that I do not observe who suffers such a shock without becoming homeless, since the reason for housing loss is recorded only inside the shelter system: while a nationwide database of such notices does not exist in France, individuals formally report the reason for their housing loss when requesting shelter. I therefore study the selected sample of individuals who enter the shelter system through this channel and leverage the quasi-random variation in the *timing* of the shock. To ensure the analysis captures the impact on active workers, I restrict the sample to individuals with recent labor market attachment, defined as having been employed for at least six months within the two years preceding the shock. This results in a final analytical sample of 2,757 individuals.

Specifically, I follow [Callaway and Sant’Anna \(2021\)](#); [Sant’Anna and Zhao \(2020\)](#) and estimate

$$Y_{it} = \delta_t + \alpha_i + \beta \text{HousingLoss}_{it} + \epsilon_{it}, \quad (\text{E13})$$

where Y_{it} denotes the outcome variable for individual i in year t and HousingLoss_{it} is a binary indicator equal to one in the year an individual first requests shelter due to a property recla-

mation and in all subsequent periods. The control group consists of not-yet-treated units. All specifications incorporate individual and year fixed effects. Standard errors are clustered at the individual level and estimated using a multiplicative wild bootstrap.

The identifying assumption is that the timing of the property reclamation is quasi-random: in the absence of the housing shock, individuals in the different treatment cohorts would have followed parallel labor market trajectories. Under this assumption, any divergence in outcomes after the shock can be attributed to the timing of the housing loss rather than to systematic differences in prior labor market attachment.

Threats to identification. Expand, per notes. Points to cover: (i) the estimand is the effect of housing loss among those who eventually request shelter after a landlord repossession, not among all households receiving such a notice; (ii) anticipation — the notice period (six months for a *congé pour vente*) means workers may adjust before the request date, which would show up as a pre-trend; the flat pre-trends argue against this; (iii) cohort composition — later cohorts may differ (e.g. post-COVID rental market); (iv) the design should be most powerful in tight rental markets, notably Paris, which cannot yet be tested given the sample size.

E.2 Results

[Table E1](#) reports the aggregate estimates, pooling all pre- and post-event periods. Columns (1) and (2) examine two employment margins: an indicator for any formal employment, including very short-term contracts, and a restricted measure for employment lasting at least one month. In both specifications, I observe a stark decline, 32% for any employment and 30% for more stable employment, relative to the not-yet-treated control group. Earnings contract by 32% relative to the baseline mean. Pre-treatment averages are economically negligible and provide no evidence of differential trends.

[Figure E1](#) and [Figure E2](#) show that the effects of housing loss worsen over time. By the first post-event year, the employment effect is roughly –15 percentage points; by the second year, –21 percentage points; and by the fourth year, approximately –33 percentage points. The pattern is more pronounced for male workers (right panels), for whom the year-4 employment effect approaches –40 percentage points. Earnings deteriorate in parallel, from a decline of €170 in year 1 to €420 in year 4 (–€520 in year 4 for men).

This pattern contrasts with the eviction literature, in which labor market outcomes display a stark *pre-event* dip that partially recovers post-eviction ([Collinson et al., 2024](#)). Here, pre-trends

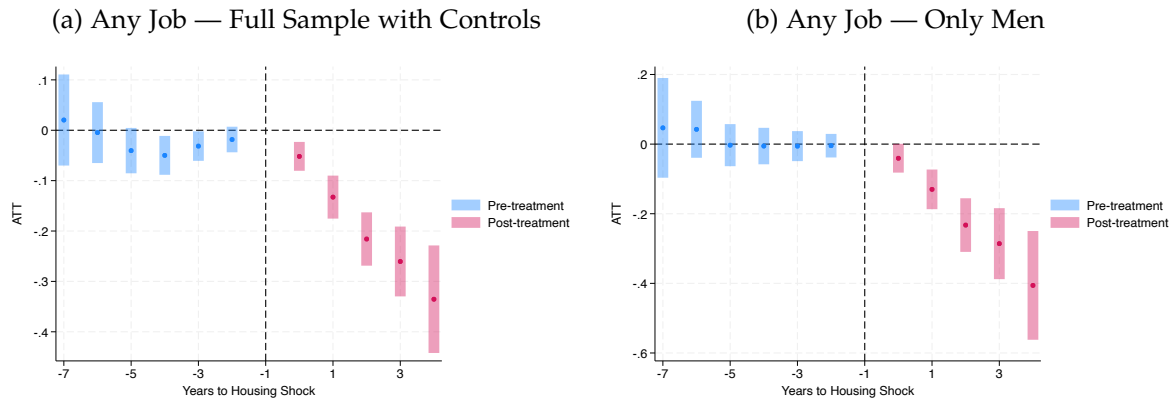
are flat and the post-event divergence accelerates over time, consistent with housing instability eroding labor market attachment progressively. For the “any employment” and earnings measures, I detect a minor pre-trend, though its magnitude is economically negligible; when restricting the analysis to stable employment (contracts exceeding one month), no such pre-trend is observed.

Table E1: The Effect of Housing Loss on Labor Market Trajectories

	Any Job	Job $\geq$ 1 months	Earnings
	(1)	(2)	(3)
Effect of housing loss	-0.20*** (0.017)	-0.16*** (0.018)	-246.14*** (20.66)
Pre-treatment average	-0.02 (0.003)	-0.02 (0.003)	-24.0 (4.05)
Mean outcome, treated at t= 0	0.62	0.53	781.11
Effect relative to mean	-32%	-30%	-32%
Number of obs.	21,536	21,536	21,536
Controls: Gender & Age Groups	Yes	Yes	Yes

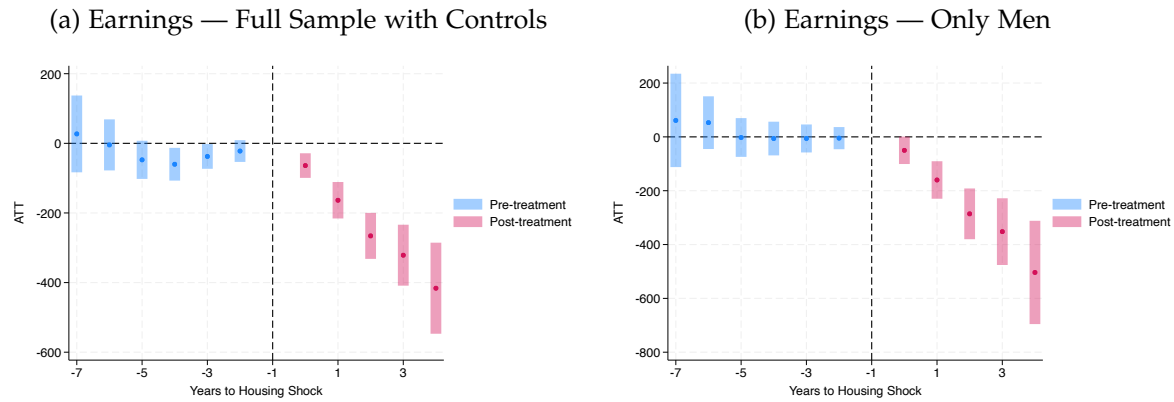
Notes: This table reports the estimated impact of a residential shock on labor market trajectories using the doubly robust estimator proposed by [Sant'Anna and Zhao \(2020\)](#) and [Callaway and Sant'Anna \(2021\)](#). The identification strategy exploits staggered treatment timing, utilizing 'not-yet-treated' individuals as the control group for those experiencing a property reclamation in a given period. All specifications incorporate individual and year fixed effects. Standard errors are clustered at the individual level and estimated via a multiplicative Wild Bootstrap procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure E1: The Effect of Housing Loss on Employment Worsens Over Time



Notes: This figure illustrates the dynamic impact of a residential shock on employment, estimated using the doubly robust framework of [Sant'Anna and Zhao \(2020\)](#) and [Callaway and Sant'Anna \(2021\)](#). The identification strategy exploits staggered treatment timing by utilizing 'not-yet-treated' individuals as a control group for those experiencing property reclamation in earlier periods. Shaded areas denote 95% confidence intervals, calculated via a multiplicative Wild Bootstrap procedure, with standard errors clustered at the individual level. All specifications incorporate individual and year fixed effects. The left panel shows the full sample with gender and age controls; the right panel restricts to male workers. The horizontal axis is years relative to the housing shock.

Figure E2: The Effect of Housing Loss on Earnings Worsens Over Time



Notes: Dynamic ATT estimates on monthly earnings (euros). The estimator is from [Sant'Anna and Zhao \(2020\)](#) and [Callaway and Sant'Anna \(2021\)](#), exploiting staggered treatment timing with 'not-yet-treated' individuals as controls. Shaded areas denote 95% confidence intervals, calculated via multiplicative Wild Bootstrap, standard errors clustered at the individual level. All specifications include individual and year fixed effects. Left panel: full sample with gender and age controls. Right panel: male workers only.